Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Bar Mac Dairy

Physical address of dairy:

Number and Street	City	County	Zip Code
5070 S Hunt RD	Gustine	Merced	95322

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 06/01/2003

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X063-X230-X001-XXXX

B. OPERATORS

Nightengale, Brad

Operator name: Nightengale, Brad

Telephone no.: (209) 854-1273 Landline, (209) 617-3087 Cellular

Mailing Address Number and Street	City	State	Zip Code
5070 S Hunt RD	Gustine	CA	95322

This operator is responsible for paying permit fees.

C. OWNERS

Nightengale, Brad

Legal owner name: Nightengale, Brad

Telephone no.: (209) 854-1273 Landline, (209) 617-3087 Cellular

Mailing Address Number and Street	City	State	Zip Code
5070 S Hunt RD	Gustine	CA	95322

This owner is responsible for paying permit fees.

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	930	0	0	0
Number under roof	2,065	310	0	0	0	0
Maximum number	2,100	315	1,000	0	0	0
Average number	2,065	310	930	0	0	0
Avg live weight (lbs)	1,400	1,400	900	0		

Predominant milk cow breed: Holstein

Average milk production: 70 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 66,853.68 *tons per reporting period*

Total nitrogen from manure: 837,391.91 *lbs per reporting period*

After ammonia losses (30% loss applied): 586,174.34 *lbs per reporting period*

Total phosphorus from manure: 139,649.93 *lbs per reporting period*

Total potassium from manure: 395,466.55 *lbs per reporting period*

Total salt from manure: 1,043,589.75 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 32,400,000 *gallons*

Total nitrogen generated: 172,120.38 *lbs*

Total phosphorus generated: 20,972.73 *lbs*

Total potassium generated: 202,981.40 *lbs*

Total salt generated: 1,713,047.41 *lbs*

	32,400,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	32,400,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
CCID Canal	Surface water
Drain SW	Surface water
Irrigation Wells	Ground water
IW 24	Ground water
IW 7	Ground water

Source Description	Type
IW Hunt	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

Date	Material type / Description	Quantity	Reporting basis	Moisture (%)	N (%)	P (%)	K (%)	Salt (%)
01/06/2023	Solid commercial fertilizer *Urea*	5.65 *ton*	As-is	0.1	46.000000	0.000000	0.000000	0.000000
06/12/2023	Solid commercial fertilizer *7-17-3*	11.63 *ton*	As-is	0.1	7.000000	17.000000	3.000000	0.000000
07/01/2023	Solid commercial fertilizer *UN 32*	24.02 *ton*	As-is	0.1	32.000000	0.000000	0.000000	0.000000

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Commercial fertilizer / Other	22,199.00	3,954.20	697.80	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Total imports for all materials	22,199.00	3,954.20	697.80	0.00

G. NUTRIENT EXPORTS

Date	Material type	Quantity	Reporting basis	Moisture (%)	Density (lbs/cu ft)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/28/2023	Separator solids	3,960.00 *ton*	Dry-weight	74.7		19,400.00	3,160.00	7,390.00		0.00
05/29/2023	Corral solids	4,650.00 *ton*	Dry-weight	66.0		14,500.00	5,900.00	30,900.00		67.10
12/16/2023	Separator solids	6,600.00 *ton*	Dry-weight	73.6		17,000.00	2,420.00	4,970.00		67.30

No liquid nutrient exports entered.

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Dry manure	143,963.54	33,420.90	129,833.04	4,466,972.40
Process wastewater	0.00	0.00	0.00	0.00
Total exports for all materials	143,963.54	33,420.90	129,833.04	4,466,972.40

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APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Field 1	47	47	2	process wastewater	X063-X230-X003-XXXX
Field 10	43	43	1	none	X063-X230-X001-XXXX X063-X230-X014-XXXX X063-X230-X015-XXXX
Field 14	48	48	2	both	X063-X230-X003-XXXX
Field 18/19	67	67	2	process wastewater	X063-X230-X010-XXXX
Field 2	90	90	2	both	X063-X230-X003-XXXX X063-X230-X016-XXXX X063-X230-X017-XXXX
Field 20	42	42	2	process wastewater	X063-X230-X010-XXXX
Field 21	54	54	2	process wastewater	X063-X230-X010-XXXX
Field 22	32	32	1	process wastewater	X063-X230-X007-XXXX
Field 23	34	34	2	process wastewater	X063-X230-X007-XXXX
Field 24	33	33	1	manure	X063-X230-X007-XXXX
Field 25	18	18	1	none	X063-X230-X007-XXXX
Field 26	19	19	1	manure	X063-X230-X007-XXXX
Field 27	18	18	1	process wastewater	X063-X230-X007-XXXX
Field 28&29	15	15	2	none	X063-X230-X007-XXXX
Field 3	64	64	2	process wastewater	X063-X230-X003-XXXX X063-X230-X016-XXXX
Field 30	80	80	0	none	X063-X230-X008-XXXX
Field 6	48	48	2	process wastewater	X063-X230-X001-XXXX
Field 7	67	67	2	process wastewater	X063-X230-X001-XXXX X063-X230-X015-XXXX
Totals for areas that were used for application	663	663	24		
Totals for areas that were not used for application	156	156	4		
Land application area totals	819	819	28		

B. CROPS AND HARVESTS

Field 1

Field name: Field 1

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Field 1

10/20/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 47 Plant date: 10/20/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/17/2023	900.95 *ton*	As-is		69.9	6,000.00	900.00	7,800.00		10.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.00	187.60	29.40	116.20	0.00
Total actual harvest content	19.17	230.03	34.50	299.04	1,234.76

06/19/2023: Corn, silage

Crop: Corn, silage Acres planted: 47 Plant date: 06/19/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/24/2023	1,006.59 *ton*	As-is		71.1	3,900.00	600.00	300.00		4.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	20.00	204.00	40.00	210.00	0.00
Total actual harvest content	21.42	167.05	25.70	12.85	569.43

Field 10

Field name: Field 10

10/21/2021: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 43 Plant date: 10/21/2021

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/12/2023	322.07 *ton*	As-is		7.5	33,700.00	3,600.00	20,700.00		10.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	480.00	43.20	336.00	0.00
Total actual harvest content	7.49	504.83	53.93	310.09	1,510.36

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Field 14

Field name: Field 14

10/25/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 48 Plant date: 10/25/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/14/2023	945.55 *ton*	As-is		70.7	6,400.00	1,100.00	9,300.00		10.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.00	187.60	29.40	116.20	0.00
Total actual harvest content	19.70	252.15	43.34	366.40	1,258.25

06/20/2023: Corn, silage

Crop: Corn, silage Acres planted: 48 Plant date: 06/20/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/28/2023	976.30 *ton*	As-is		67.3	3,300.00	700.00	4,100.00		4.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	20.00	204.00	40.00	210.00	0.00
Total actual harvest content	20.34	134.24	28.48	166.78	545.39

Field 18/19

Field name: Field 18/19

10/03/2022: Rye Grass Silage

Crop: Rye Grass Silage Acres planted: 67 Plant date: 10/03/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
06/02/2023	1,070.00 *ton*	As-is		56.4	10,500.00	1,600.00	7,400.00		11.15

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	214.40	33.60	228.80	0.00
Total actual harvest content	15.97	335.37	51.10	236.36	1,552.75

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Field 18/19

07/26/2023: Milo Silage

Crop: Milo Silage Acres planted: 67 Plant date: 07/26/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/02/2023	220.00 *ton*	As-is		48.9	9,500.00	1,200.00	13,400.00		10.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	153.00	30.00	157.50	0.00
Total actual harvest content	3.28	62.39	7.88	88.00	342.29

Field 2

Field name: Field 2

10/27/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 90 Plant date: 10/27/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/15/2023	1,552.69 *ton*	As-is		66.1	3,900.00	700.00	6,700.00		9.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.00	187.60	29.40	116.20	0.00
Total actual harvest content	17.25	134.57	24.15	231.18	1,052.72

06/17/2023: Corn, silage

Crop: Corn, silage Acres planted: 90 Plant date: 06/17/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/27/2023	1,378.87 *ton*	As-is		72.4	3,200.00	600.00	4,000.00		5.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	20.00	204.00	40.00	210.00	0.00
Total actual harvest content	15.32	98.05	18.38	122.57	456.68

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Field 20

Field name: Field 20

10/03/2022: Rye Grass Silage

Crop: Rye Grass Silage Acres planted: 42 Plant date: 10/03/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
06/12/2023	819.00 *ton*	As-is		64.5	9,800.00	1,000.00	9,100.00		12.55

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	214.40	33.60	228.80	0.00
Total actual harvest content	19.50	382.20	39.00	354.90	1,737.55

07/28/2023: Milo Silage

Crop: Milo Silage Acres planted: 42 Plant date: 07/28/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/02/2023	286.00 *ton*	As-is		58.5	5,200.00	800.00	8,500.00		8.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	153.00	30.00	157.50	0.00
Total actual harvest content	6.81	70.82	10.90	115.76	497.37

Field 21

Field name: Field 21

11/01/2002: Pasture

Crop: Pasture Acres planted: 54 Plant date: 11/01/2002

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/14/2023	701.00 *ton*	As-is		66.9	7,700.00	1,000.00	9,700.00		12.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	226.50	31.50	265.50	0.00
Total actual harvest content	12.98	199.91	25.96	251.84	1,065.62

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Field 21

10/26/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 54 Plant date: 10/26/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/13/2023	767.29 *ton*	As-is		60.8	4,500.00	900.00	8,100.00		9.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.00	187.60	29.40	116.20	0.00
Total actual harvest content	14.21	127.88	25.58	230.19	1,091.71

Field 22

Field name: Field 22

11/01/2002: Pasture

Crop: Pasture Acres planted: 32 Plant date: 11/01/2002

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/13/2023	659.00 *ton*	As-is		63.8	8,300.00	1,100.00	10,000.00		9.83

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	226.50	31.50	265.50	0.00
Total actual harvest content	20.59	341.86	45.31	411.88	1,465.64

Field 23

Field name: Field 23

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Field 23

10/01/2022: Rye Grass Silage

Crop: Rye Grass Silage Acres planted: 34 Plant date: 10/01/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
06/12/2023	575.00 *ton*	As-is		58.1	8,500.00	2,900.00	2,800.00		15.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	214.40	33.60	228.80	0.00
Total actual harvest content	16.91	287.50	98.09	94.71	2,253.36

07/21/2023: Milo Silage

Crop: Milo Silage Acres planted: 34 Plant date: 07/21/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/23/2023	90.00 *ton*	As-is		48.9	9,500.00	1,200.00	13,400.00		10.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	153.00	30.00	157.50	0.00
Total actual harvest content	2.65	50.29	6.35	70.94	275.94

Field 24

Field name: Field 24

11/01/2022: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 33 Plant date: 11/01/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/14/2023	174.03 *ton*	As-is		10.5	35,700.00	3,100.00	19,000.00		11.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	480.00	43.20	336.00	0.00
Total actual harvest content	5.27	376.54	32.70	200.40	1,057.26

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Field 25

Field name: Field 25

10/21/2021: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 18 Plant date: 10/21/2021

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/13/2023	124.78 *ton*	As-is		9.9	29,200.00	3,400.00	19,400.00		10.95

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	480.00	43.20	336.00	0.00
Total actual harvest content	6.93	404.84	47.14	268.97	1,367.86

Field 26

Field name: Field 26

11/03/2022: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 19 Plant date: 11/03/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/14/2023	133.31 *ton*	As-is		20.8	29,000.00	2,500.00	16,800.00		11.35

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	480.00	43.20	336.00	0.00
Total actual harvest content	7.02	406.95	35.08	235.75	1,261.42

Field 27

Field name: Field 27

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Field 27

11/01/2002: Pasture

Crop: Pasture Acres planted: 18 Plant date: 11/01/2002

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/11/2023	449.00 *ton*	As-is		64.6	7,900.00	1,000.00	9,900.00		10.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	226.50	31.50	265.50	0.00
Total actual harvest content	24.94	394.12	49.89	493.90	1,819.05

Field 28&29

Field name: Field 28&29

10/01/2022: Rye Grass Silage

Crop: Rye Grass Silage Acres planted: 15 Plant date: 10/01/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
06/12/2023	168.00 *ton*	As-is		56.3	8,100.00	800.00	2,800.00		18.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	214.40	33.60	228.80	0.00
Total actual harvest content	11.20	181.44	17.92	62.72	1,781.56

07/25/2023: Milo Silage

Crop: Milo Silage Acres planted: 15 Plant date: 07/25/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/23/2023	30.00 *ton*	As-is		48.9	9,500.00	1,200.00	13,400.00		10.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	153.00	30.00	157.50	0.00
Total actual harvest content	2.00	38.00	4.80	53.60	208.49

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Field 3

Field name: Field 3

10/21/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 64 Plant date: 10/21/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/15/2023	1,031.67 *ton*	As-is		69.6	7,000.00	1,100.00	9,400.00		10.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.00	187.60	29.40	116.20	0.00
Total actual harvest content	16.12	225.68	35.46	303.05	980.09

06/16/2023: Corn, silage

Crop: Corn, silage Acres planted: 64 Plant date: 06/16/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/27/2023	1,090.38 *ton*	As-is		67.4	3,800.00	700.00	3,700.00		4.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	20.00	204.00	40.00	210.00	0.00
Total actual harvest content	17.04	129.48	23.85	126.08	466.55

Field 6

Field name: Field 6

10/18/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 48 Plant date: 10/18/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/13/2023	895.52 *ton*	As-is		69.8	7,200.00	1,200.00	9,900.00		11.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.00	187.60	29.40	116.20	0.00
Total actual harvest content	18.66	268.66	44.78	369.40	1,262.09

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Field 6

06/12/2023: Corn, silage

Crop: Corn, silage Acres planted: 48 Plant date: 06/12/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/12/2023	707.06 *ton*	As-is		68.7	3,900.00	700.00	4,600.00		6.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	20.00	204.00	40.00	210.00	0.00
Total actual harvest content	14.73	114.90	20.62	135.52	562.50

Field 7

Field name: Field 7

10/19/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 67 Plant date: 10/19/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/17/2023	1,075.06 *ton*	As-is		64.4	3,700.00	800.00	6,700.00		9.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.00	187.60	29.40	116.20	0.00
Total actual harvest content	16.05	118.74	25.67	215.01	1,062.48

06/14/2023: Corn, silage

Crop: Corn, silage Acres planted: 67 Plant date: 06/14/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/13/2023	1,076.57 *ton*	As-is		67.1	4,200.00	800.00	4,800.00		5.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	20.00	204.00	40.00	210.00	0.00
Total actual harvest content	16.07	134.97	25.71	154.25	592.08

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NUTRIENT BUDGET

A. LAND APPLICATIONS

Field 1 - 10/20/2022: Wheat, silage, soft dough

Field name: Field 1

Crop: Wheat, silage, soft dough — Plant date: 10/20/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	271.02	34.20	240.34	1,751.38	1,800,000.00 *gal*
CCID Canal	Surface water	1.52	0.00	0.00	171.81	6,451,200.00 *gal*
Application event totals		272.54	34.20	240.34	1,923.20	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.18	0.00	0.00	133.16	4,999,680.00 *gal*
Application event totals		1.18	0.00	0.00	133.16	

Field 1 - 06/19/2023: Corn, silage

Field name: Field 1

Crop: Corn, silage — Plant date: 06/19/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/19/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
7-17-3	Solid commercial fertilizer	4.47	10.86	1.92	0.00	
Application event totals		4.47	10.86	1.92	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/03/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	42.24	0.00	0.00	0.00	
Application event totals		42.24	0.00	0.00	0.00	

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Field 1 - 06/19/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	7.28	0.00	0.00	1,584.81	9,108,000.00 *gal*
CCID Canal	Surface water	1.66	0.00	0.00	186.85	7,015,680.00 *gal*
Application event totals		8.93	0.00	0.00	1,771.66	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	5.96	0.00	0.00	1,297.71	7,458,000.00 *gal*
Application event totals		5.96	0.00	0.00	1,297.71	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	120.23	14.00	222.27	1,418.53	1,485,000.00 *gal*
Irrigation Wells	Ground water	5.48	0.00	0.00	1,194.35	6,864,000.00 *gal*
Application event totals		125.72	14.00	222.27	2,612.88	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	6.33	0.00	0.00	1,378.10	7,920,000.00 *gal*
Application event totals		6.33	0.00	0.00	1,378.10	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	5.06	0.00	0.00	1,102.48	6,336,000.00 *gal*
Application event totals		5.06	0.00	0.00	1,102.48	

Field 10 - 10/21/2021: Alfalfa, hay

Field name: Field 10

Crop: Alfalfa, hay

Plant date: 10/21/2021

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Field 10 - 10/21/2021: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/15/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Drain SW	Surface water	4.41	0.00	0.00	643.54	5,472,000.00 *gal*
Application event totals		4.41	0.00	0.00	643.54	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Drain SW	Surface water	5.33	0.00	0.00	777.61	6,612,000.00 *gal*
Application event totals		5.33	0.00	0.00	777.61	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Drain SW	Surface water	5.05	0.00	0.00	737.39	6,270,000.00 *gal*
Application event totals		5.05	0.00	0.00	737.39	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Drain SW	Surface water	4.96	0.00	0.00	723.98	6,156,000.00 *gal*
Application event totals		4.96	0.00	0.00	723.98	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Drain SW	Surface water	4.68	0.00	0.00	683.76	5,814,000.00 *gal*
Application event totals		4.68	0.00	0.00	683.76	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Drain SW	Surface water	4.96	0.00	0.00	723.98	6,156,000.00 *gal*
Application event totals		4.96	0.00	0.00	723.98	

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Field 10 - 10/21/2021: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Drain SW	Surface water	4.32	0.00	0.00	630.13	5,358,000.00 *gal*
Application event totals		4.32	0.00	0.00	630.13	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Drain SW	Surface water	3.76	0.00	0.00	549.69	4,674,000.00 *gal*
Application event totals		3.76	0.00	0.00	549.69	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Drain SW	Surface water	5.05	0.00	0.00	737.39	6,270,000.00 *gal*
Application event totals		5.05	0.00	0.00	737.39	

Field 14 - 10/25/2022: Wheat, silage, soft dough

Field name: Field 14

Crop: Wheat, silage, soft dough

Plant date: 10/25/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/03/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	180.55	44.08	276.62	2,289.28	320.00 *ton*
Application event totals		180.55	44.08	276.62	2,289.28	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/01/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Drain SW	Surface water	6.83	0.00	0.00	996.87	9,462,000.00 *gal*
Application event totals		6.83	0.00	0.00	996.87	

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Field 14 - 10/25/2022: Wheat, silage, soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Drain SW	Surface water	3.29	0.00	0.00	480.42	4,560,000.00 *gal*
Application event totals		3.29	0.00	0.00	480.42	

03/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	139.32	17.58	123.55	900.32	945,000.00 *gal*
Drain SW	Surface water	6.58	0.00	0.00	960.84	9,120,000.00 *gal*
Application event totals		145.90	17.58	123.55	1,861.16	

Field 14 - 06/20/2023: Corn, silage

Field name: Field 14

Crop: Corn, silage

Plant date: 06/20/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	2.11	0.00	0.00	237.63	9,112,320.00 *gal*
Application event totals		2.11	0.00	0.00	237.63	

06/20/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
7-17-3	Solid commercial fertilizer	4.47	10.86	1.92	0.00	
Application event totals		4.47	10.86	1.92	0.00	

07/04/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	42.24	0.00	0.00	0.00	
Application event totals		42.24	0.00	0.00	0.00	

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Field 14 - 06/20/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	4.65	0.00	0.00	1,012.04	5,940,000.00 *gal*
Application event totals		4.65	0.00	0.00	1,012.04	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	5.16	0.00	0.00	1,124.49	6,600,000.00 *gal*
Application event totals		5.16	0.00	0.00	1,124.49	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	96.32	11.22	178.07	1,136.43	1,215,000.00 *gal*
Irrigation Wells	Ground water	4.85	0.00	0.00	1,057.02	6,204,000.00 *gal*
Application event totals		101.18	11.22	178.07	2,193.45	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	4.96	0.00	0.00	1,079.51	6,336,000.00 *gal*
Application event totals		4.96	0.00	0.00	1,079.51	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	5.68	0.00	0.00	1,236.94	7,260,000.00 *gal*
Application event totals		5.68	0.00	0.00	1,236.94	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	4.96	0.00	0.00	1,079.51	6,336,000.00 *gal*
Application event totals		4.96	0.00	0.00	1,079.51	

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Field 18/19 - 10/03/2022: Rye Grass Silage

Field name: Field 18/19

Crop: Rye Grass Silage

Plant date: 10/03/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/10/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	185.24	25.98	87.44	3,418.96	2,250,000.00 *gal*
Drain SW	Surface water	10.19	0.00	0.00	1,488.59	19,722,000.00 *gal*
Application event totals		195.43	25.98	87.44	4,907.55	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	151.91	11.68	259.89	2,554.92	2,520,000.00 *gal*
Drain SW	Surface water	8.31	0.00	0.00	1,213.24	16,074,000.00 *gal*
Application event totals		160.22	11.68	259.89	3,768.16	

Field 18/19 - 07/26/2023: Milo Silage

Field name: Field 18/19

Crop: Milo Silage

Plant date: 07/26/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	0.94	0.00	0.00	134.65	7,560,000.00 *gal*
Drain SW	Surface water	5.30	0.00	0.00	774.41	10,260,000.00 *gal*
Application event totals		6.24	0.00	0.00	909.06	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Drain SW	Surface water	4.83	0.00	0.00	705.57	9,348,000.00 *gal*
Application event totals		4.83	0.00	0.00	705.57	

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Field 18/19 - 07/26/2023: Milo Silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Drain SW	Surface water	6.60	0.00	0.00	963.71	12,768,000.00 *gal*
Application event totals		6.60	0.00	0.00	963.71	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Drain SW	Surface water	4.89	0.00	0.00	714.18	9,462,000.00 *gal*
Application event totals		4.89	0.00	0.00	714.18	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Drain SW	Surface water	4.18	0.00	0.00	610.92	8,094,000.00 *gal*
Application event totals		4.18	0.00	0.00	610.92	

Field 2 - 10/27/2022: Wheat, silage, soft dough

Field name: Field 2

Crop: Wheat, silage, soft dough

Plant date: 10/27/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/01/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	163.70	39.97	250.80	2,075.61	544.00 *ton*
Application event totals		163.70	39.97	250.80	2,075.61	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/30/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.94	0.00	0.00	218.71	15,724,800.00 *gal*
Application event totals		1.94	0.00	0.00	218.71	

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Field 2 - 10/27/2022: Wheat, silage, soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Drain SW	Surface water	3.07	0.00	0.00	448.39	7,980,000.00 *gal*
Application event totals		3.07	0.00	0.00	448.39	

Field 2 - 06/17/2023: Corn, silage

Field name: Field 2

Crop: Corn, silage

Plant date: 06/17/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	2.15	0.00	0.00	243.01	17,472,000.00 *gal*
Application event totals		2.15	0.00	0.00	243.01	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/17/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
7-17-3	Solid commercial fertilizer	4.47	10.86	1.92	0.00	
Application event totals		4.47	10.86	1.92	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	0.46	0.00	0.00	65.63	4,950,000.00 *gal*
Application event totals		0.46	0.00	0.00	65.63	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/05/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	42.24	0.00	0.00	0.00	
Application event totals		42.24	0.00	0.00	0.00	

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Field 2 - 06/17/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.59	0.00	0.00	179.45	12,902,400.00 *gal*
Application event totals		1.59	0.00	0.00	179.45	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Drain SW	Surface water	5.61	0.00	0.00	819.92	14,592,000.00 *gal*
Application event totals		5.61	0.00	0.00	819.92	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	78.01	9.08	144.21	920.37	1,845,000.00 *gal*
Drain SW	Surface water	6.32	0.00	0.00	922.41	16,416,000.00 *gal*
Application event totals		84.33	9.08	144.21	1,842.78	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	0.23	0.00	0.00	33.41	2,520,000.00 *gal*
Drain SW	Surface water	5.40	0.00	0.00	787.89	14,022,000.00 *gal*
Application event totals		5.63	0.00	0.00	821.30	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Drain SW	Surface water	5.57	0.00	0.00	813.51	14,478,000.00 *gal*
Application event totals		5.57	0.00	0.00	813.51	

Field 20 - 10/03/2022: Rye Grass Silage

Field name: Field 20

Crop: Rye Grass Silage

Plant date: 10/03/2022

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Field 20 - 10/03/2022: Rye Grass Silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/20/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Drain SW	Surface water	7.24	0.00	0.00	1,056.93	8,778,000.00 *gal*
Application event totals		7.24	0.00	0.00	1,056.93	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	303.28	38.27	268.95	1,959.88	1,800,000.00 *gal*
Drain SW	Surface water	6.58	0.00	0.00	960.84	7,980,000.00 *gal*
Application event totals		309.86	38.27	268.95	2,920.73	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Drain SW	Surface water	6.20	0.00	0.00	905.94	7,524,000.00 *gal*
Application event totals		6.20	0.00	0.00	905.94	

Field 20 - 07/28/2023: Milo Silage

Field name: Field 20

Crop: Milo Silage

Plant date: 07/28/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	1.41	0.00	0.00	202.01	7,110,000.00 *gal*
Application event totals		1.41	0.00	0.00	202.01	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Drain SW	Surface water	6.39	0.00	0.00	933.39	7,752,000.00 *gal*
Application event totals		6.39	0.00	0.00	933.39	

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Field 20 - 07/28/2023: Milo Silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Drain SW	Surface water	5.36	0.00	0.00	782.40	6,498,000.00 *gal*
Application event totals		5.36	0.00	0.00	782.40	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	1.45	0.00	0.00	207.13	7,290,000.00 *gal*
Application event totals		1.45	0.00	0.00	207.13	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Drain SW	Surface water	5.83	0.00	0.00	851.03	7,068,000.00 *gal*
Application event totals		5.83	0.00	0.00	851.03	

Field 21 - 11/01/2002: Pasture

Field name: Field 21

Crop: Pasture

Plant date: 11/01/2002

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	1.00	0.00	0.00	143.20	6,480,000.00 *gal*
Application event totals		1.00	0.00	0.00	143.20	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	228.32	26.59	422.09	2,693.77	3,240,000.00 *gal*
Drain SW	Surface water	8.41	0.00	0.00	1,227.74	13,110,000.00 *gal*
Application event totals		236.73	26.59	422.09	3,921.51	

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Field 21 - 11/01/2002: Pasture

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Drain SW	Surface water	4.02	0.00	0.00	587.18	6,270,000.00 *gal*
Application event totals		4.02	0.00	0.00	587.18	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Drain SW	Surface water	5.78	0.00	0.00	843.41	9,006,000.00 *gal*
Application event totals		5.78	0.00	0.00	843.41	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Drain SW	Surface water	4.61	0.00	0.00	672.59	7,182,000.00 *gal*
Application event totals		4.61	0.00	0.00	672.59	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Drain SW	Surface water	3.58	0.00	0.00	523.13	5,586,000.00 *gal*
Application event totals		3.58	0.00	0.00	523.13	

Field 21 - 10/26/2022: Wheat, silage, soft dough

Field name: Field 21

Crop: Wheat, silage, soft dough Plant date: 10/26/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/16/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Drain SW	Surface water	8.70	0.00	0.00	1,270.45	13,566,000.00 *gal*
Application event totals		8.70	0.00	0.00	1,270.45	

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Field 21 - 10/26/2022: Wheat, silage, soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	147.43	18.60	130.74	952.72	1,125,000.00 *gal*
Drain SW	Surface water	6.51	0.00	0.00	950.17	10,146,000.00 *gal*
Application event totals		153.94	18.60	130.74	1,902.89	

Field 22 - 11/01/2002: Pasture

Field name: Field 22

Crop: Pasture

Plant date: 11/01/2002

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	169.17	21.35	150.02	1,093.25	765,000.00 *gal*
Drain SW	Surface water	3.70	0.00	0.00	540.47	3,420,000.00 *gal*
Application event totals		172.88	21.35	150.02	1,633.72	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Drain SW	Surface water	4.81	0.00	0.00	702.62	4,446,000.00 *gal*
Application event totals		4.81	0.00	0.00	702.62	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Drain SW	Surface water	5.43	0.00	0.00	792.70	5,016,000.00 *gal*
Application event totals		5.43	0.00	0.00	792.70	

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Field 22 - 11/01/2002: Pasture

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Drain SW	Surface water	5.92	0.00	0.00	864.76	5,472,000.00 *gal*
Application event totals		5.92	0.00	0.00	864.76	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Drain SW	Surface water	3.95	0.00	0.00	576.51	3,648,000.00 *gal*
Application event totals		3.95	0.00	0.00	576.51	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Drain SW	Surface water	3.45	0.00	0.00	504.44	3,192,000.00 *gal*
Application event totals		3.45	0.00	0.00	504.44	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	107.02	12.46	197.85	1,262.70	900,000.00 *gal*
Drain SW	Surface water	4.32	0.00	0.00	630.55	3,990,000.00 *gal*
Application event totals		111.34	12.46	197.85	1,893.26	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Drain SW	Surface water	3.21	0.00	0.00	468.41	2,964,000.00 *gal*
Application event totals		3.21	0.00	0.00	468.41	

Field 23 - 10/01/2022: Rye Grass Silage

Field name: Field 23

Crop: Rye Grass Silage

Plant date: 10/01/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

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Field 23 - 10/01/2022: Rye Grass Silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/13/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Drain SW	Surface water	6.73	0.00	0.00	983.45	6,612,000.00 *gal*
Application event totals		6.73	0.00	0.00	983.45	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	280.98	35.45	249.17	1,815.77	1,350,000.00 *gal*
Drain SW	Surface water	3.48	0.00	0.00	508.68	3,420,000.00 *gal*
Application event totals		284.46	35.45	249.17	2,324.46	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	0.91	0.00	0.00	129.51	3,690,000.00 *gal*
Drain SW	Surface water	4.76	0.00	0.00	695.20	4,674,000.00 *gal*
Application event totals		5.67	0.00	0.00	824.71	

Field 23 - 07/21/2023: Milo Silage

Field name: Field 23

Crop: Milo Silage

Plant date: 07/21/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.26	0.00	0.00	142.51	3,870,720.00 *gal*
Drain SW	Surface water	5.57	0.00	0.00	813.89	5,472,000.00 *gal*
Application event totals		6.84	0.00	0.00	956.40	

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Field 23 - 07/21/2023: Milo Silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Drain SW	Surface water	4.06	0.00	0.00	593.46	3,990,000.00 *gal*
Application event totals		4.06	0.00	0.00	593.46	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Drain SW	Surface water	4.18	0.00	0.00	610.42	4,104,000.00 *gal*
Application event totals		4.18	0.00	0.00	610.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Drain SW	Surface water	4.99	0.00	0.00	729.11	4,902,000.00 *gal*
Application event totals		4.99	0.00	0.00	729.11	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Drain SW	Surface water	4.64	0.00	0.00	678.24	4,560,000.00 *gal*
Application event totals		4.64	0.00	0.00	678.24	

Field 24 - 11/01/2022: Alfalfa, hay

Field name: Field 24

Crop: Alfalfa, hay

Plant date: 11/01/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/26/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	246.21	60.11	377.21	3,121.75	300.00 *ton*
Application event totals		246.21	60.11	377.21	3,121.75	

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Field 24 - 11/01/2022: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	1.21	0.00	0.00	172.74	4,777,000.00 *gal*
Application event totals		1.21	0.00	0.00	172.74	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	1.32	0.00	0.00	188.76	5,220,000.00 *gal*
Application event totals		1.32	0.00	0.00	188.76	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	1.43	0.00	0.00	205.04	5,670,000.00 *gal*
Application event totals		1.43	0.00	0.00	205.04	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	1.02	0.00	0.00	146.45	4,050,000.00 *gal*
Application event totals		1.02	0.00	0.00	146.45	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	1.09	0.00	0.00	156.22	4,320,000.00 *gal*
Application event totals		1.09	0.00	0.00	156.22	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	1.37	0.00	0.00	195.27	5,400,000.00 *gal*
Application event totals		1.37	0.00	0.00	195.27	

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Field 24 - 11/01/2022: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	1.14	0.00	0.00	162.73	4,500,000.00 *gal*
Application event totals		1.14	0.00	0.00	162.73	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	1.18	0.00	0.00	169.24	4,680,000.00 *gal*
Application event totals		1.18	0.00	0.00	169.24	

Field 25 - 10/21/2021: Alfalfa, hay

Field name: Field 25

Crop: Alfalfa, hay

Plant date: 10/21/2021

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/19/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Drain SW	Surface water	4.61	0.00	0.00	672.59	2,394,000.00 *gal*
Application event totals		4.61	0.00	0.00	672.59	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Drain SW	Surface water	5.70	0.00	0.00	832.73	2,964,000.00 *gal*
Application event totals		5.70	0.00	0.00	832.73	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	1.42	0.00	0.00	202.87	3,060,000.00 *gal*
Application event totals		1.42	0.00	0.00	202.87	

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Field 25 - 10/21/2021: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	1.00	0.00	0.00	143.20	2,160,000.00 *gal*
Application event totals		1.00	0.00	0.00	143.20	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	1.08	0.00	0.00	155.13	2,340,000.00 *gal*
Application event totals		1.08	0.00	0.00	155.13	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	1.21	0.00	0.00	173.03	2,610,000.00 *gal*
Application event totals		1.21	0.00	0.00	173.03	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Drain SW	Surface water	5.48	0.00	0.00	800.70	2,850,000.00 *gal*
Application event totals		5.48	0.00	0.00	800.70	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Drain SW	Surface water	5.70	0.00	0.00	832.73	2,964,000.00 *gal*
Application event totals		5.70	0.00	0.00	832.73	

Field 26 - 11/03/2022: Alfalfa, hay

Field name: Field 26

Crop: Alfalfa, hay

Plant date: 11/03/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

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Field 26 - 11/03/2022: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/28/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	242.32	59.16	371.25	3,072.45	170.00 *ton*
Application event totals		242.32	59.16	371.25	3,072.45	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	0.67	0.00	0.00	96.09	1,530,000.00 *gal*
Drain SW	Surface water	3.32	0.00	0.00	485.48	1,824,000.00 *gal*
Application event totals		4.00	0.00	0.00	581.57	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Drain SW	Surface water	7.69	0.00	0.00	1,122.67	4,218,000.00 *gal*
Application event totals		7.69	0.00	0.00	1,122.67	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.93	0.00	0.00	217.82	3,306,240.00 *gal*
Application event totals		1.93	0.00	0.00	217.82	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	1.54	0.00	0.00	220.45	3,510,000.00 *gal*
Application event totals		1.54	0.00	0.00	220.45	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	1.66	0.00	0.00	237.41	3,780,000.00 *gal*
Application event totals		1.66	0.00	0.00	237.41	

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Field 26 - 11/03/2022: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	1.30	0.00	0.00	186.54	2,970,000.00 *gal*
Application event totals		1.30	0.00	0.00	186.54	

Field 27 - 11/01/2002: Pasture

Field name: Field 27

Crop: Pasture

Plant date: 11/01/2002

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	159.22	20.09	141.20	1,028.94	405,000.00 *gal*
Drain SW	Surface water	6.58	0.00	0.00	960.84	3,420,000.00 *gal*
Application event totals		165.80	20.09	141.20	1,989.78	

05/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Drain SW	Surface water	4.17	0.00	0.00	608.53	2,166,000.00 *gal*
Application event totals		4.17	0.00	0.00	608.53	

06/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Drain SW	Surface water	4.83	0.00	0.00	704.62	2,508,000.00 *gal*
Application event totals		4.83	0.00	0.00	704.62	

07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Drain SW	Surface water	5.26	0.00	0.00	768.67	2,736,000.00 *gal*
Application event totals		5.26	0.00	0.00	768.67	

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Field 27 - 11/01/2002: Pasture

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Drain SW	Surface water	4.61	0.00	0.00	672.59	2,394,000.00 *gal*
Application event totals		4.61	0.00	0.00	672.59	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	142.70	16.62	263.81	1,683.60	675,000.00 *gal*
Drain SW	Surface water	5.04	0.00	0.00	736.65	2,622,000.00 *gal*
Application event totals		147.74	16.62	263.81	2,420.25	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Drain SW	Surface water	3.95	0.00	0.00	576.51	2,052,000.00 *gal*
Application event totals		3.95	0.00	0.00	576.51	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Drain SW	Surface water	3.29	0.00	0.00	480.42	1,710,000.00 *gal*
Application event totals		3.29	0.00	0.00	480.42	

Field 28&29 - 10/01/2022: Rye Grass Silage

Field name: Field 28&29

Crop: Rye Grass Silage

Plant date: 10/01/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/06/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Urea	Solid commercial fertilizer	150.00	0.00	0.00	0.00	
Application event totals		150.00	0.00	0.00	0.00	

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Field 28&29 - 10/01/2022: Rye Grass Silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Drain SW	Surface water	5.66	0.00	0.00	826.33	2,451,000.00 *gal*
Application event totals		5.66	0.00	0.00	826.33	

04/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Drain SW	Surface water	8.95	0.00	0.00	1,306.75	3,876,000.00 *gal*
Application event totals		8.95	0.00	0.00	1,306.75	

Field 28&29 - 07/25/2023: Milo Silage

Field name: Field 28&29

Crop: Milo Silage

Plant date: 07/25/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Drain SW	Surface water	7.24	0.00	0.00	1,056.93	3,135,000.00 *gal*
Application event totals		7.24	0.00	0.00	1,056.93	

08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Drain SW	Surface water	4.87	0.00	0.00	711.02	2,109,000.00 *gal*
Application event totals		4.87	0.00	0.00	711.02	

08/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Drain SW	Surface water	6.58	0.00	0.00	960.84	2,850,000.00 *gal*
Application event totals		6.58	0.00	0.00	960.84	

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Field 28&29 - 07/25/2023: Milo Silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Drain SW	Surface water	6.19	0.00	0.00	903.19	2,679,000.00 *gal*
Application event totals		6.19	0.00	0.00	903.19	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Drain SW	Surface water	5.66	0.00	0.00	826.33	2,451,000.00 *gal*
Application event totals		5.66	0.00	0.00	826.33	

Field 3 - 10/21/2022: Wheat, silage, soft dough

Field name: Field 3

Crop: Wheat, silage, soft dough

Plant date: 10/21/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/06/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Urea	Solid commercial fertilizer	46.00	0.00	0.00	0.00	
Application event totals		46.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	248.79	31.39	220.62	1,607.72	2,250,000.00 *gal*
CCID Canal	Surface water	1.12	0.00	0.00	126.18	6,451,200.00 *gal*
Application event totals		249.90	31.39	220.62	1,733.89	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.55	0.00	0.00	175.07	8,951,040.00 *gal*
Application event totals		1.55	0.00	0.00	175.07	

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Field 3 - 06/16/2023: Corn, silage

Field name: Field 3

Crop: Corn, silage

Plant date: 06/16/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.86	0.00	0.00	209.77	10,725,120.00 *gal*
Application event totals		1.86	0.00	0.00	209.77	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/16/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
7-17-3	Solid commercial fertilizer	4.47	10.86	1.92	0.00	
Application event totals		4.47	10.86	1.92	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/02/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	42.24	0.00	0.00	0.00	
Application event totals		42.24	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	0.50	0.00	0.00	56.78	2,903,040.00 *gal*
Drain SW	Surface water	3.70	0.00	0.00	540.47	6,840,000.00 *gal*
Application event totals		4.20	0.00	0.00	597.25	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	4.76	0.00	0.00	1,037.34	8,118,000.00 *gal*
Application event totals		4.76	0.00	0.00	1,037.34	

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Field 3 - 06/16/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	85.62	9.97	158.28	1,010.16	1,440,000.00 *gal*
Drain SW	Surface water	4.38	0.00	0.00	639.56	8,094,000.00 *gal*
Application event totals		90.00	9.97	158.28	1,649.72	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Drain SW	Surface water	5.31	0.00	0.00	774.68	9,804,000.00 *gal*
Application event totals		5.31	0.00	0.00	774.68	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	0.69	0.00	0.00	77.28	3,951,360.00 *gal*
Drain SW	Surface water	5.86	0.00	0.00	855.75	10,830,000.00 *gal*
Application event totals		6.55	0.00	0.00	933.03	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Drain SW	Surface water	4.44	0.00	0.00	648.57	8,208,000.00 *gal*
Application event totals		4.44	0.00	0.00	648.57	

Field 6 - 10/18/2022: Wheat, silage, soft dough

Field name: Field 6

Crop: Wheat, silage, soft dough

Plant date: 10/18/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

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Field 6 - 10/18/2022: Wheat, silage, soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/25/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	124.11	17.41	58.58	2,290.70	1,080,000.00 *gal*
CCID Canal	Surface water	2.72	0.00	0.00	307.03	11,773,440.00 *gal*
Application event totals		126.83	17.41	58.58	2,597.73	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	179.13	22.60	158.85	1,157.56	1,215,000.00 *gal*
CCID Canal	Surface water	1.49	0.00	0.00	168.24	6,451,200.00 *gal*
Application event totals		180.62	22.60	158.85	1,325.79	

Field 6 - 06/12/2023: Corn, silage

Field name: Field 6

Crop: Corn, silage

Plant date: 06/12/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	2.24	0.00	0.00	252.35	9,676,800.00 *gal*
Application event totals		2.24	0.00	0.00	252.35	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/12/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
7-17-3	Solid commercial fertilizer	4.47	10.86	1.92	0.00	
Application event totals		4.47	10.86	1.92	0.00	

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Field 6 - 06/12/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/01/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	42.24	0.00	0.00	0.00	
Application event totals		42.24	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.49	0.00	0.00	168.24	6,451,200.00 *gal*
Application event totals		1.49	0.00	0.00	168.24	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	2.44	0.00	0.00	274.78	10,536,960.00 *gal*
Application event totals		2.44	0.00	0.00	274.78	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	96.32	11.22	178.07	1,136.43	1,215,000.00 *gal*
Drain SW	Surface water	8.72	0.00	0.00	1,273.12	12,084,000.00 *gal*
Application event totals		105.04	11.22	178.07	2,409.55	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	11.47	0.00	0.00	1,306.36	6,534,000.00 *gal*
Application event totals		11.47	0.00	0.00	1,306.36	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.34	0.00	0.00	151.41	5,806,080.00 *gal*
Application event totals		1.34	0.00	0.00	151.41	

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Field 7 - 10/19/2022: Wheat, silage, soft dough

Field name: Field 7

Crop: Wheat, silage, soft dough

Plant date: 10/19/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/25/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	142.59	17.99	126.45	921.44	1,350,000.00 *gal*
CCID Canal	Surface water	1.92	0.00	0.00	216.95	11,612,160.00 *gal*
Application event totals		144.51	17.99	126.45	1,138.39	

Field 7 - 06/14/2023: Corn, silage

Field name: Field 7

Crop: Corn, silage

Plant date: 06/14/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	2.95	0.00	0.00	332.96	17,821,440.00 *gal*
Application event totals		2.95	0.00	0.00	332.96	

06/14/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
7-17-3	Solid commercial fertilizer	4.47	10.86	1.92	0.00	
Application event totals		4.47	10.86	1.92	0.00	

07/01/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	42.24	0.00	0.00	0.00	
Application event totals		42.24	0.00	0.00	0.00	

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Field 7 - 06/14/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.60	0.00	0.00	180.79	9,676,800.00 *gal*
Application event totals		1.60	0.00	0.00	180.79	

07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	1.35	0.00	0.00	192.36	10,800,000.00 *gal*
Application event totals		1.35	0.00	0.00	192.36	

08/05/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	86.90	10.12	160.65	1,025.24	1,530,000.00 *gal*
Drain SW	Surface water	6.31	0.00	0.00	920.69	12,198,000.00 *gal*
Application event totals		93.20	10.12	160.65	1,945.93	

08/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	1.43	0.00	0.00	205.18	11,520,000.00 *gal*
Application event totals		1.43	0.00	0.00	205.18	

09/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Drain SW	Surface water	3.54	0.00	0.00	516.27	6,840,000.00 *gal*
Application event totals		3.54	0.00	0.00	516.27	

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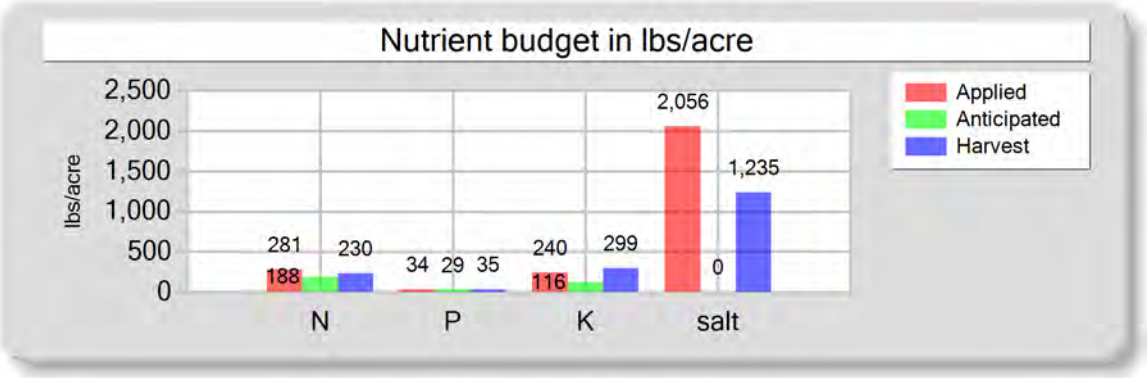
B. NUTRIENT BUDGET

Field 1 - 10/20/2022: Wheat, silage, soft dough

Field name: Field 1 Crop: Wheat, silage, soft dough Plant date: 10/20/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	271.02	34.20	240.34	1,751.38
Fresh water	2.70	0.00	0.00	304.97
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	280.72	34.20	240.34	2,056.36
Anticipated crop nutrient removal	187.60	29.40	116.20	0.00
Actual crop nutrient removal	230.03	34.50	299.04	1,234.76
Nutrient balance	50.69	-0.31	-58.70	821.59
Applied to removed ratio	1.22	0.99	0.80	1.67

Fresh water applied	
11,450,880.00	*gallons*
421.70	*acre-inches*
8.97	*inches/acre*

Process wastewater applied	
1,800,000.00	*gallons*
66.29	*acre-inches*
1.41	*inches/acre*

Total harvests for the crop	
1	*harvests*

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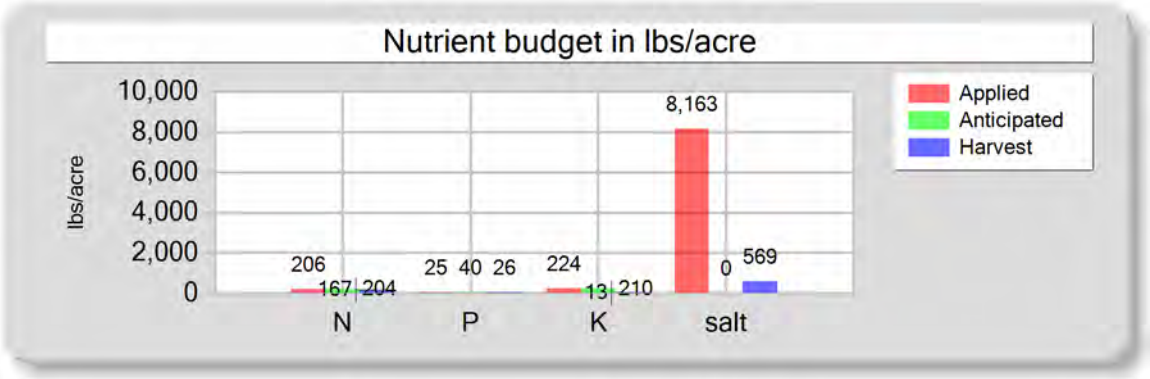
Field 1 - 06/19/2023: Corn, silage

Field name: Field 1 Crop: Corn, silage Plant date: 06/19/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	46.71	10.86	1.92	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	120.23	14.00	222.27	1,418.53
Fresh water	31.77	0.00	0.00	6,744.29
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	205.71	24.86	224.19	8,162.82
Anticipated crop nutrient removal	204.00	40.00	210.00	0.00
Actual crop nutrient removal	167.05	25.70	12.85	569.43
Nutrient balance	38.66	-0.84	211.34	7,593.39
Applied to removed ratio	1.23	0.97	17.45	14.34

Fresh water applied	
44,701,680.00	*gallons*
1,646.21	*acre-inches*
35.03	*inches/acre*

Process wastewater applied	
1,485,000.00	*gallons*
54.69	*acre-inches*
1.16	*inches/acre*

Total harvests for the crop	
1	*harvests*

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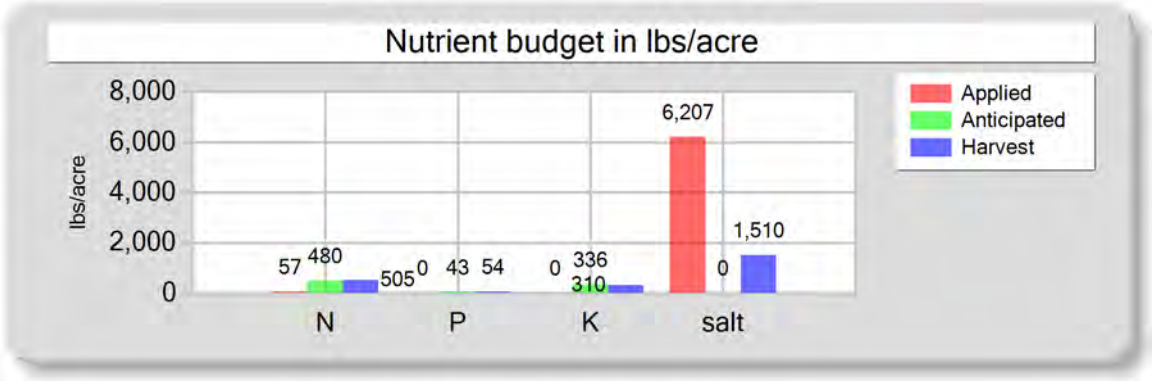
Field 10 - 10/21/2021: Alfalfa, hay

Field name: Field 10 Crop: Alfalfa, hay Plant date: 10/21/2021



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	42.51	0.00	0.00	6,207.49
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	56.51	0.00	0.00	6,207.49
Anticipated crop nutrient removal	480.00	43.20	336.00	0.00
Actual crop nutrient removal	504.83	53.93	310.09	1,510.36
Nutrient balance	-448.32	-53.93	-310.09	4,697.14
Applied to removed ratio	0.11	0.00	0.00	4.11

Fresh water applied	
52,782,000.00	*gallons*
1,943.78	*acre-inches*
45.20	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

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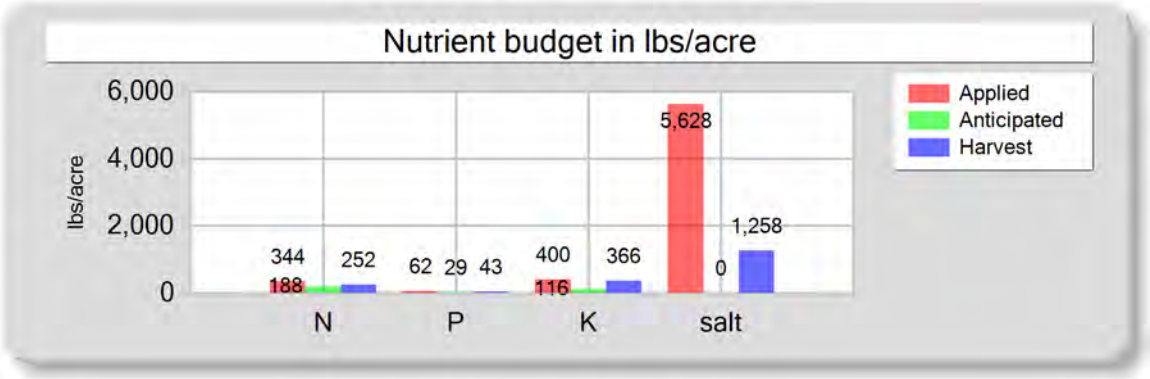
Field 14 - 10/25/2022: Wheat, silage, soft dough

Field name: Field 14 Crop: Wheat, silage, soft dough Plant date: 10/25/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	180.55	44.08	276.62	2,289.28
Process wastewater	139.32	17.58	123.55	900.32
Fresh water	16.70	0.00	0.00	2,438.14
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	343.57	61.66	400.17	5,627.74
Anticipated crop nutrient removal	187.60	29.40	116.20	0.00
Actual crop nutrient removal	252.15	43.34	366.40	1,258.25
Nutrient balance	91.42	18.32	33.77	4,369.49
Applied to removed ratio	1.36	1.42	1.09	4.47

Fresh water applied	
23,142,000.00	*gallons*
852.24	*acre-inches*
17.76	*inches/acre*

Process wastewater applied	
945,000.00	*gallons*
34.80	*acre-inches*
0.73	*inches/acre*

Total harvests for the crop	
1	*harvests*

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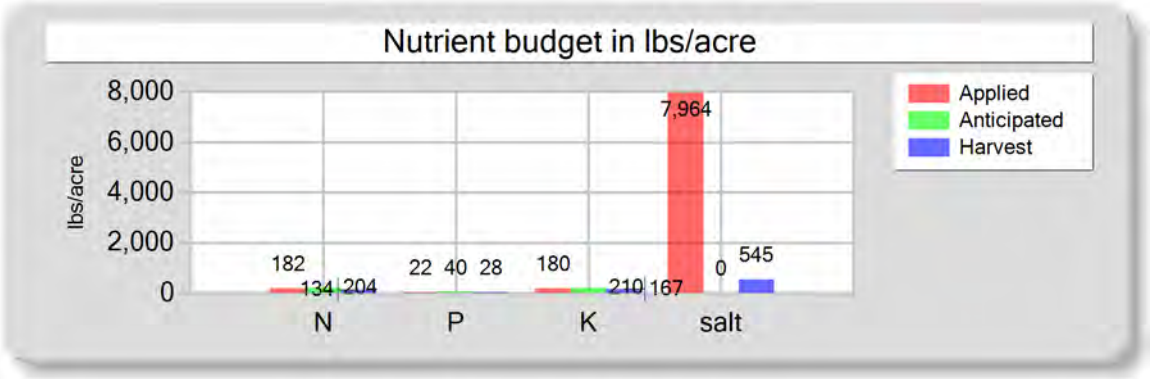
Field 14 - 06/20/2023: Corn, silage

Field name: Field 14 Crop: Corn, silage Plant date: 06/20/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	46.71	10.86	1.92	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	96.32	11.22	178.07	1,136.43
Fresh water	32.36	0.00	0.00	6,827.14
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	182.40	22.08	179.99	7,963.57
Anticipated crop nutrient removal	204.00	40.00	210.00	0.00
Actual crop nutrient removal	134.24	28.48	166.78	545.39
Nutrient balance	48.16	-6.40	13.20	7,418.18
Applied to removed ratio	1.36	0.78	1.08	14.60

Fresh water applied	
47,788,320.00	*gallons*
1,759.88	*acre-inches*
36.66	*inches/acre*

Process wastewater applied	
1,215,000.00	*gallons*
44.74	*acre-inches*
0.93	*inches/acre*

Total harvests for the crop	
1	*harvests*

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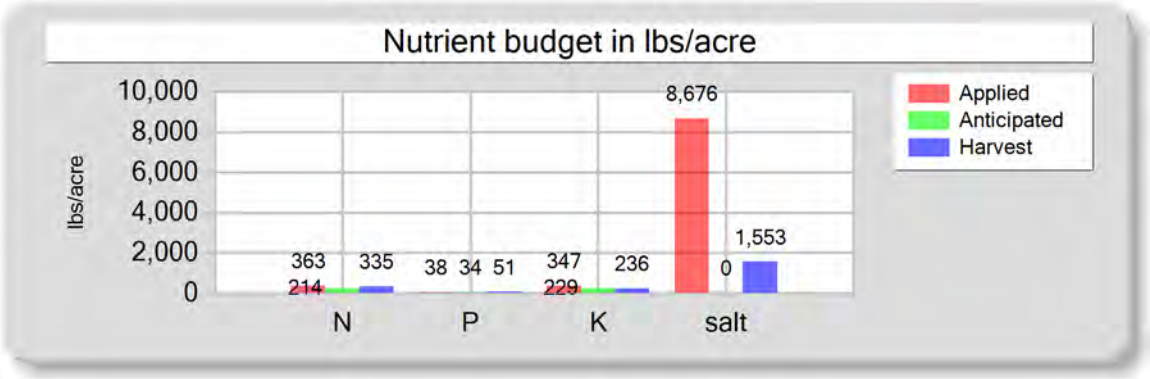
Field 18/19 - 10/03/2022: Rye Grass Silage

Field name: Field 18/19 Crop: Rye Grass Silage Plant date: 10/03/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	337.15	37.65	347.32	5,973.87
Fresh water	18.50	0.00	0.00	2,701.83
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	362.66	37.65	347.32	8,675.71
Anticipated crop nutrient removal	214.40	33.60	228.80	0.00
Actual crop nutrient removal	335.37	51.10	236.36	1,552.75
Nutrient balance	27.28	-13.45	110.96	7,122.96
Applied to removed ratio	1.08	0.74	1.47	5.59

Fresh water applied	
35,796,000.00	*gallons*
1,318.24	*acre-inches*
19.68	*inches/acre*

Process wastewater applied	
4,770,000.00	*gallons*
175.66	*acre-inches*
2.62	*inches/acre*

Total harvests for the crop	
1	*harvests*

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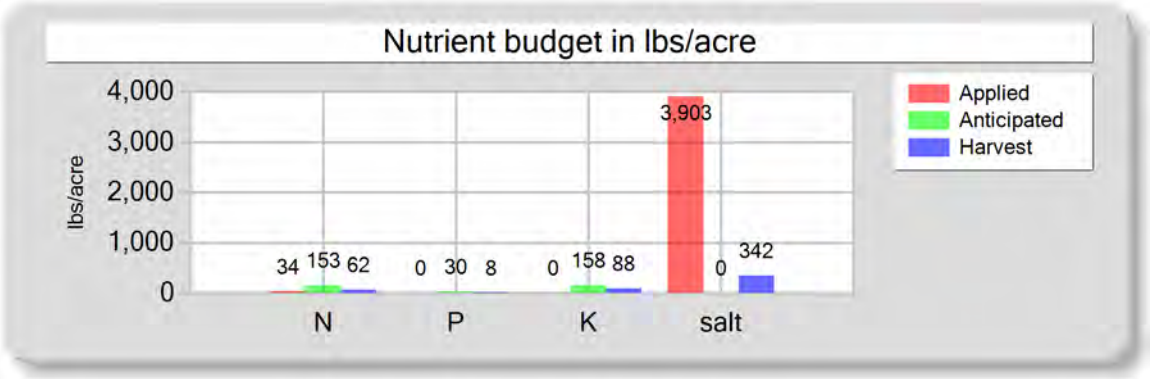
Field 18/19 - 07/26/2023: Milo Silage

Field name: Field 18/19 Crop: Milo Silage Plant date: 07/26/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	26.75	0.00	0.00	3,903.45
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	33.75	0.00	0.00	3,903.45
Anticipated crop nutrient removal	153.00	30.00	157.50	0.00
Actual crop nutrient removal	62.39	7.88	88.00	342.29
Nutrient balance	-28.64	-7.88	-88.00	3,561.16
Applied to removed ratio	0.54	0.00	0.00	11.40

Fresh water applied	
57,492,000.00	*gallons*
2,117.23	*acre-inches*
31.60	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

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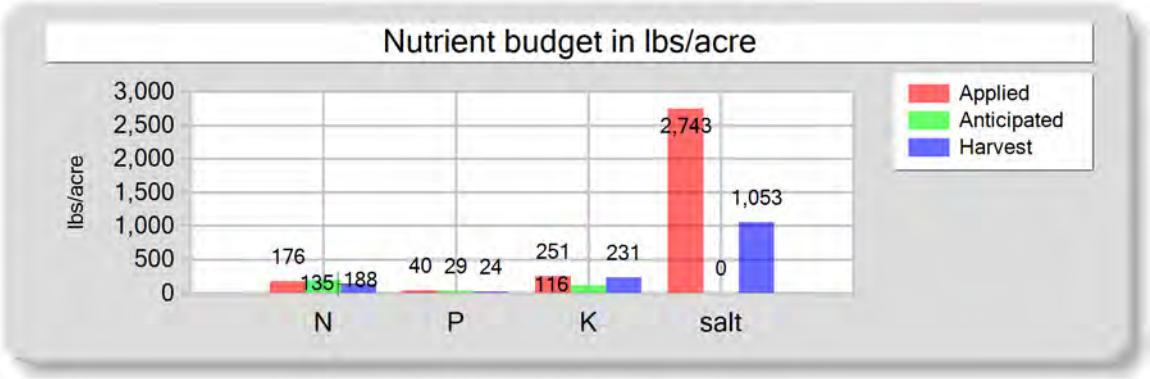
Field 2 - 10/27/2022: Wheat, silage, soft dough

Field name: Field 2 Crop: Wheat, silage, soft dough Plant date: 10/27/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	163.70	39.97	250.80	2,075.61
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	5.01	0.00	0.00	667.10
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	175.71	39.97	250.80	2,742.71
Anticipated crop nutrient removal	187.60	29.40	116.20	0.00
Actual crop nutrient removal	134.57	24.15	231.18	1,052.72
Nutrient balance	41.15	15.81	19.63	1,689.99
Applied to removed ratio	1.31	1.65	1.08	2.61

Fresh water applied	
23,704,800.00	*gallons*
872.97	*acre-inches*
9.70	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

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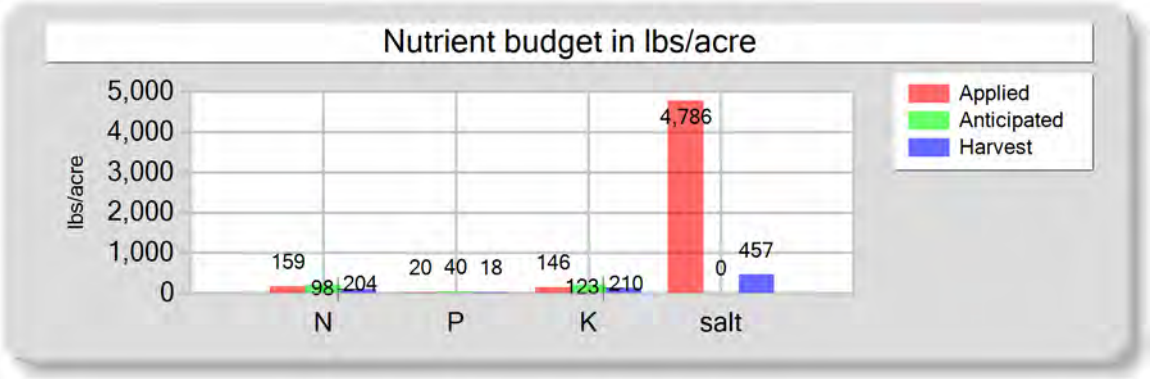
Field 2 - 06/17/2023: Corn, silage

Field name: Field 2 Crop: Corn, silage Plant date: 06/17/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	46.71	10.86	1.92	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	78.01	9.08	144.21	920.37
Fresh water	27.34	0.00	0.00	3,865.24
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	159.06	19.94	146.13	4,785.61
Anticipated crop nutrient removal	204.00	40.00	210.00	0.00
Actual crop nutrient removal	98.05	18.38	122.57	456.68
Nutrient balance	61.00	1.56	23.57	4,328.93
Applied to removed ratio	1.62	1.08	1.19	10.48

Fresh water applied	
97,352,400.00	*gallons*
3,585.16	*acre-inches*
39.84	*inches/acre*

Process wastewater applied	
1,845,000.00	*gallons*
67.95	*acre-inches*
0.75	*inches/acre*

Total harvests for the crop	
1	*harvests*

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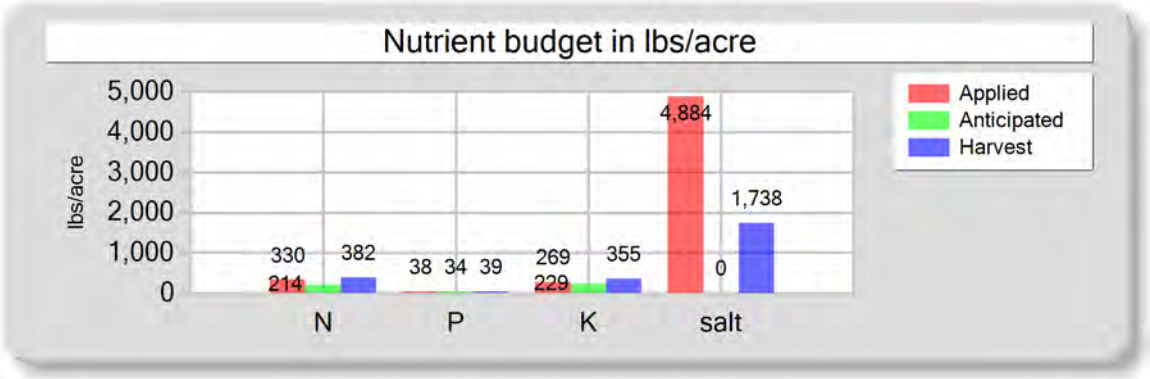
Field 20 - 10/03/2022: Rye Grass Silage

Field name: Field 20 Crop: Rye Grass Silage Plant date: 10/03/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	303.28	38.27	268.95	1,959.88
Fresh water	20.02	0.00	0.00	2,923.71
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	330.30	38.27	268.95	4,883.59
Anticipated crop nutrient removal	214.40	33.60	228.80	0.00
Actual crop nutrient removal	382.20	39.00	354.90	1,737.55
Nutrient balance	-51.90	-0.73	-85.95	3,146.04
Applied to removed ratio	0.86	0.98	0.76	2.81

Fresh water applied	
24,282,000.00	*gallons*
894.22	*acre-inches*
21.29	*inches/acre*

Process wastewater applied	
1,800,000.00	*gallons*
66.29	*acre-inches*
1.58	*inches/acre*

Total harvests for the crop	
1	*harvests*

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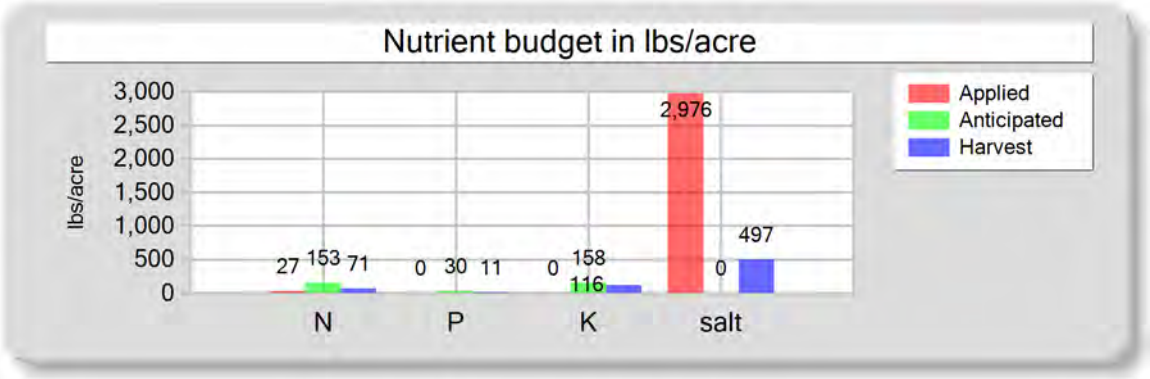
Field 20 - 07/28/2023: Milo Silage

Field name: Field 20 Crop: Milo Silage Plant date: 07/28/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	20.44	0.00	0.00	2,975.97
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	27.44	0.00	0.00	2,975.97
Anticipated crop nutrient removal	153.00	30.00	157.50	0.00
Actual crop nutrient removal	70.82	10.90	115.76	497.37
Nutrient balance	-43.38	-10.90	-115.76	2,478.60
Applied to removed ratio	0.39	0.00	0.00	5.98

Fresh water applied	
35,718,000.00	*gallons*
1,315.37	*acre-inches*
31.32	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

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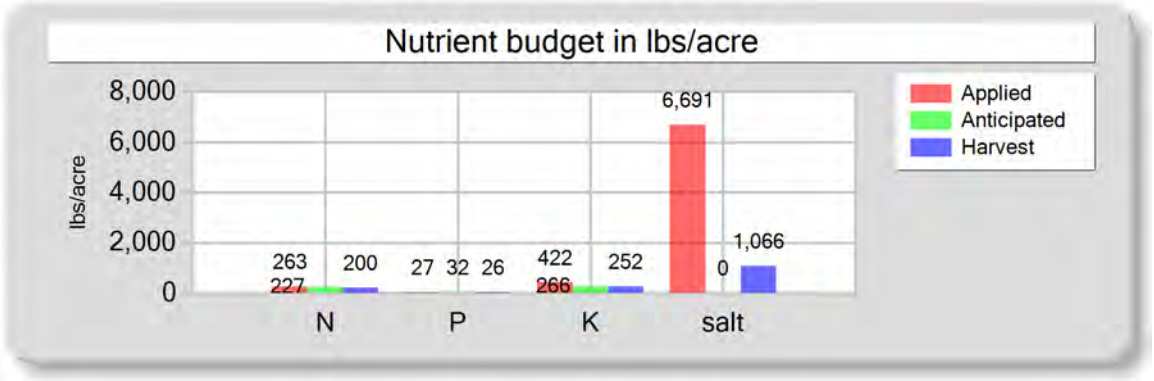
Field 21 - 11/01/2002: Pasture

Field name: Field 21 Crop: Pasture Plant date: 11/01/2002



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	228.32	26.59	422.09	2,693.77
Fresh water	27.39	0.00	0.00	3,997.25
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	262.71	26.59	422.09	6,691.02
Anticipated crop nutrient removal	226.50	31.50	265.50	0.00
Actual crop nutrient removal	199.91	25.96	251.84	1,065.62
Nutrient balance	62.80	0.62	170.25	5,625.39
Applied to removed ratio	1.31	1.02	1.68	6.28

Fresh water applied	
47,634,000.00	*gallons*
1,754.20	*acre-inches*
32.49	*inches/acre*

Process wastewater applied	
3,240,000.00	*gallons*
119.32	*acre-inches*
2.21	*inches/acre*

Total harvests for the crop	
1	*harvests*

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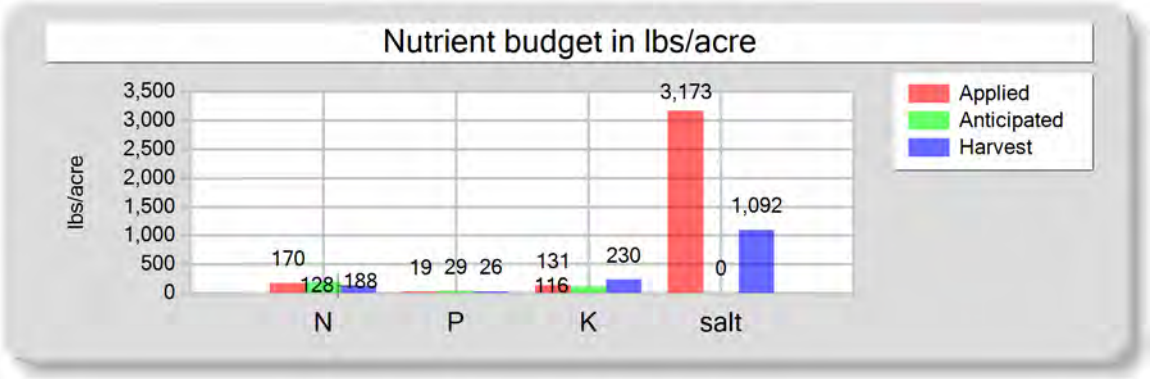
Field 21 - 10/26/2022: Wheat, silage, soft dough

Field name: Field 21 Crop: Wheat, silage, soft dough Plant date: 10/26/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	147.43	18.60	130.74	952.72
Fresh water	15.21	0.00	0.00	2,220.62
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	169.64	18.60	130.74	3,173.34
Anticipated crop nutrient removal	187.60	29.40	116.20	0.00
Actual crop nutrient removal	127.88	25.58	230.19	1,091.71
Nutrient balance	41.75	-6.97	-99.45	2,081.62
Applied to removed ratio	1.33	0.73	0.57	2.91

Fresh water applied	
23,712,000.00	*gallons*
873.23	*acre-inches*
16.17	*inches/acre*

Process wastewater applied	
1,125,000.00	*gallons*
41.43	*acre-inches*
0.77	*inches/acre*

Total harvests for the crop	
1	*harvests*

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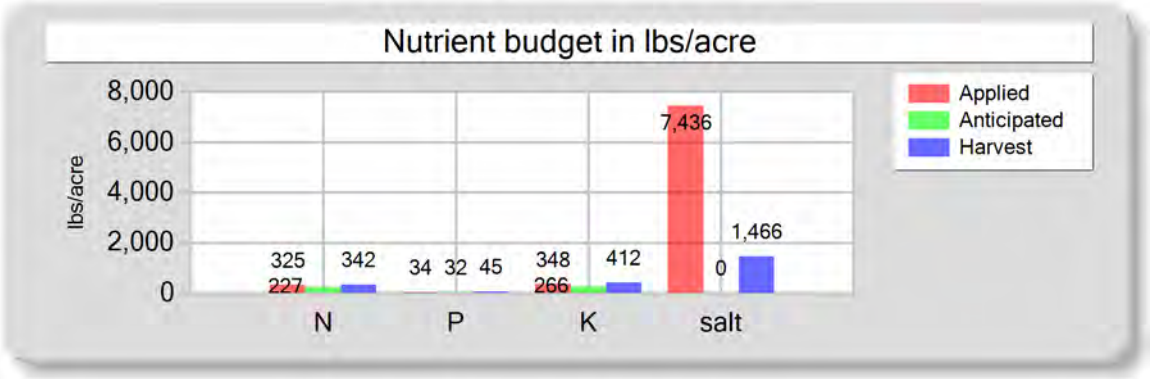
Field 22 - 11/01/2002: Pasture

Field name: Field 22 Crop: Pasture Plant date: 11/01/2002



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	276.20	33.81	347.88	2,355.95
Fresh water	34.79	0.00	0.00	5,080.46
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	324.99	33.81	347.88	7,436.41
Anticipated crop nutrient removal	226.50	31.50	265.50	0.00
Actual crop nutrient removal	341.86	45.31	411.88	1,465.64
Nutrient balance	-16.87	-11.50	-64.00	5,970.77
Applied to removed ratio	0.95	0.75	0.84	5.07

Fresh water applied	
32,148,000.00	*gallons*
1,183.90	*acre-inches*
37.00	*inches/acre*

Process wastewater applied	
1,665,000.00	*gallons*
61.32	*acre-inches*
1.92	*inches/acre*

Total harvests for the crop	
1	*harvests*

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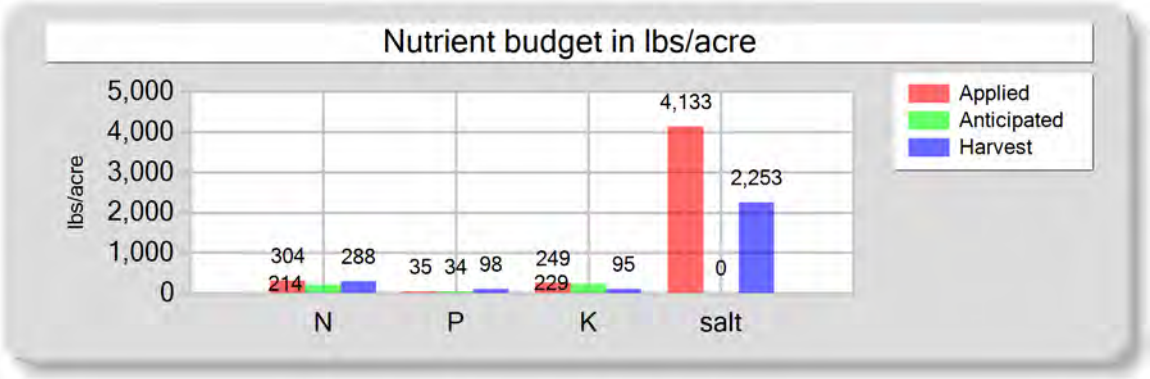
Field 23 - 10/01/2022: Rye Grass Silage

Field name: Field 23 Crop: Rye Grass Silage Plant date: 10/01/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	280.98	35.45	249.17	1,815.77
Fresh water	15.88	0.00	0.00	2,316.84
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	303.87	35.45	249.17	4,132.62
Anticipated crop nutrient removal	214.40	33.60	228.80	0.00
Actual crop nutrient removal	287.50	98.09	94.71	2,253.36
Nutrient balance	16.37	-62.63	154.47	1,879.26
Applied to removed ratio	1.06	0.36	2.63	1.83

Fresh water applied	
18,396,000.00	*gallons*
677.46	*acre-inches*
19.93	*inches/acre*

Process wastewater applied	
1,350,000.00	*gallons*
49.72	*acre-inches*
1.46	*inches/acre*

Total harvests for the crop	
1	*harvests*

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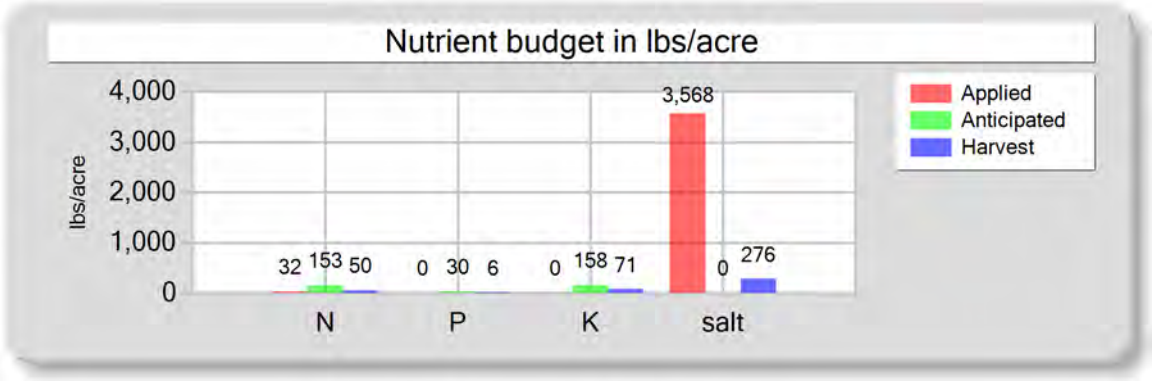
Field 23 - 07/21/2023: Milo Silage

Field name: Field 23 Crop: Milo Silage Plant date: 07/21/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	24.72	0.00	0.00	3,567.63
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	31.72	0.00	0.00	3,567.63
Anticipated crop nutrient removal	153.00	30.00	157.50	0.00
Actual crop nutrient removal	50.29	6.35	70.94	275.94
Nutrient balance	-18.57	-6.35	-70.94	3,291.69
Applied to removed ratio	0.63	0.00	0.00	12.93

Fresh water applied	
26,898,720.00	*gallons*
990.59	*acre-inches*
29.13	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

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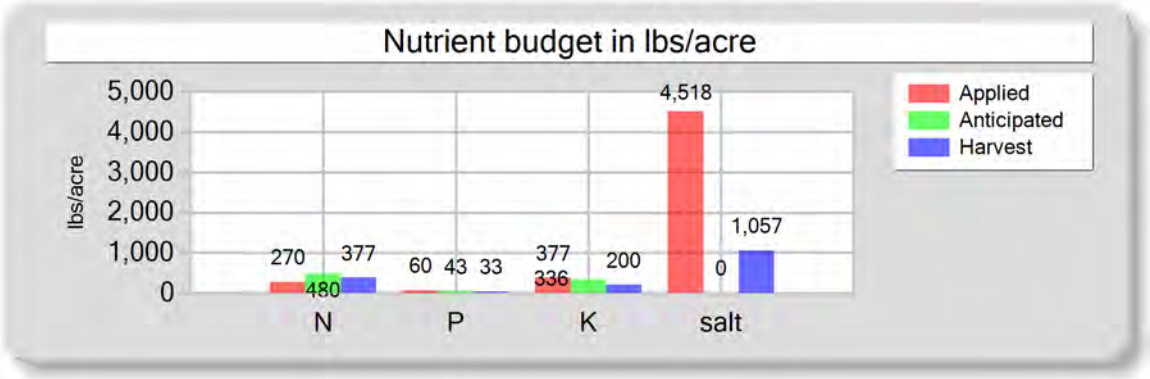
Field 24 - 11/01/2022: Alfalfa, hay

Field name: Field 24 Crop: Alfalfa, hay Plant date: 11/01/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	246.21	60.11	377.21	3,121.75
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	9.77	0.00	0.00	1,396.46
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	269.97	60.11	377.21	4,518.20
Anticipated crop nutrient removal	480.00	43.20	336.00	0.00
Actual crop nutrient removal	376.54	32.70	200.40	1,057.26
Nutrient balance	-106.56	27.42	176.81	3,460.94
Applied to removed ratio	0.72	1.84	1.88	4.27

Fresh water applied	
38,617,000.00	*gallons*
1,422.13	*acre-inches*
43.09	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

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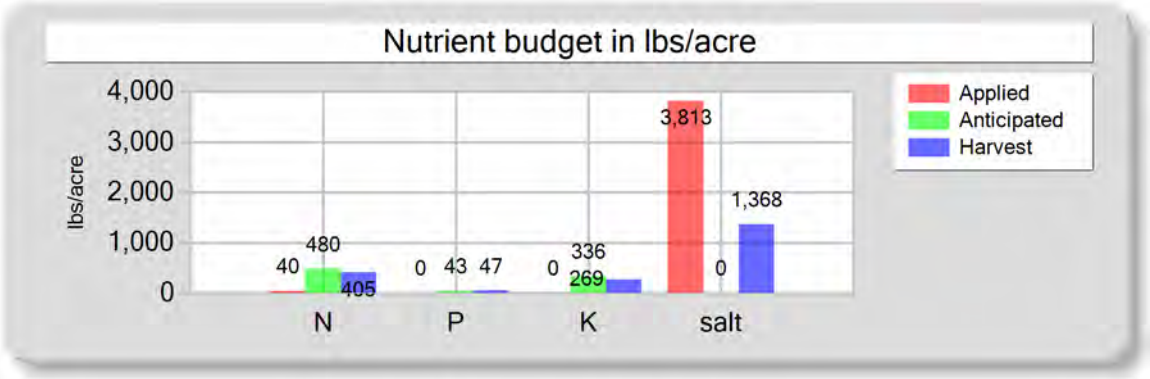
Field 25 - 10/21/2021: Alfalfa, hay

Field name: Field 25 Crop: Alfalfa, hay Plant date: 10/21/2021



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	26.21	0.00	0.00	3,812.99
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	40.21	0.00	0.00	3,812.99
Anticipated crop nutrient removal	480.00	43.20	336.00	0.00
Actual crop nutrient removal	404.84	47.14	268.97	1,367.86
Nutrient balance	-364.63	-47.14	-268.97	2,445.13
Applied to removed ratio	0.10	0.00	0.00	2.79

Fresh water applied	
21,342,000.00	*gallons*
785.95	*acre-inches*
43.66	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

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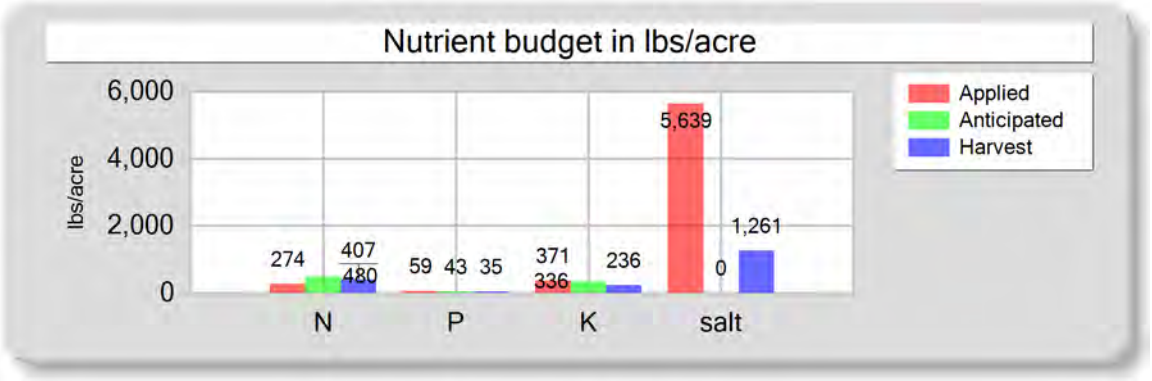
Field 26 - 11/03/2022: Alfalfa, hay

Field name: Field 26 Crop: Alfalfa, hay Plant date: 11/03/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	242.32	59.16	371.25	3,072.45
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	18.12	0.00	0.00	2,566.46
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	274.44	59.16	371.25	5,638.92
Anticipated crop nutrient removal	480.00	43.20	336.00	0.00
Actual crop nutrient removal	406.95	35.08	235.75	1,261.42
Nutrient balance	-132.50	24.08	135.51	4,377.50
Applied to removed ratio	0.67	1.69	1.57	4.47

Fresh water applied	
21,138,240.00	*gallons*
778.45	*acre-inches*
40.97	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

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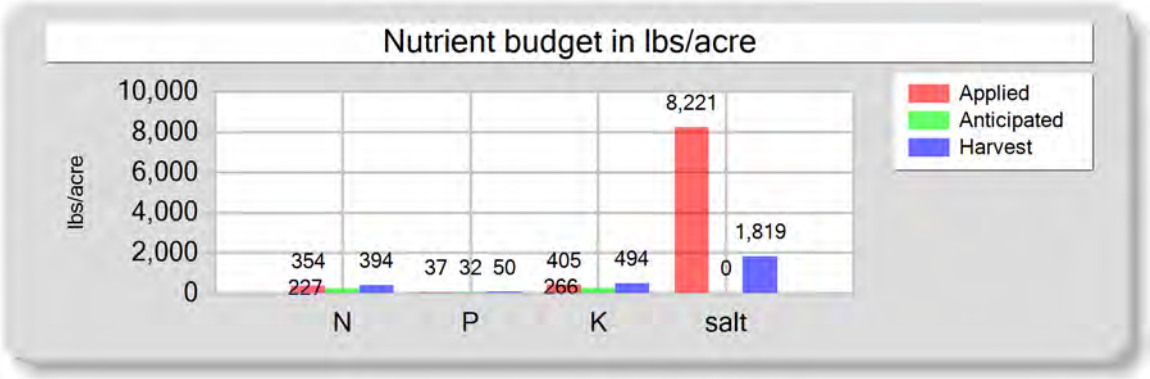
Field 27 - 11/01/2002: Pasture

Field name: Field 27 Crop: Pasture Plant date: 11/01/2002



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	301.92	36.71	405.00	2,712.54
Fresh water	37.73	0.00	0.00	5,508.83
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	353.65	36.71	405.00	8,221.38
Anticipated crop nutrient removal	226.50	31.50	265.50	0.00
Actual crop nutrient removal	394.12	49.89	493.90	1,819.05
Nutrient balance	-40.47	-13.18	-88.90	6,402.33
Applied to removed ratio	0.90	0.74	0.82	4.52

Fresh water applied	
19,608,000.00	*gallons*
722.10	*acre-inches*
40.12	*inches/acre*

Process wastewater applied	
1,080,000.00	*gallons*
39.77	*acre-inches*
2.21	*inches/acre*

Total harvests for the crop	
1	*harvests*

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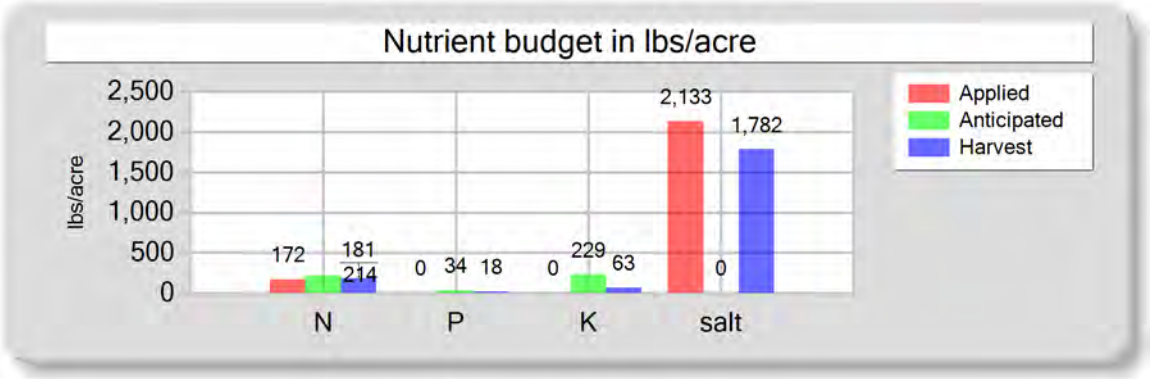
Field 28&29 - 10/01/2022: Rye Grass Silage

Field name: Field 28&29 Crop: Rye Grass Silage Plant date: 10/01/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	150.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	14.61	0.00	0.00	2,133.07
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	171.61	0.00	0.00	2,133.07
Anticipated crop nutrient removal	214.40	33.60	228.80	0.00
Actual crop nutrient removal	181.44	17.92	62.72	1,781.56
Nutrient balance	-9.83	-17.92	-62.72	351.51
Applied to removed ratio	0.95	0.00	0.00	1.20

Fresh water applied	
6,327,000.00	*gallons*
233.00	*acre-inches*
15.53	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

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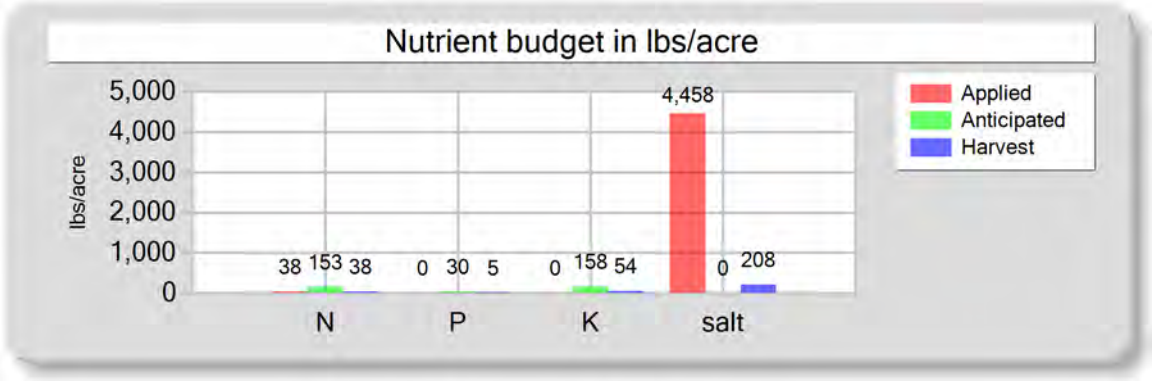
Field 28&29 - 07/25/2023: Milo Silage

Field name: Field 28&29 Crop: Milo Silage Plant date: 07/25/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	30.53	0.00	0.00	4,458.31
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	37.53	0.00	0.00	4,458.31
Anticipated crop nutrient removal	153.00	30.00	157.50	0.00
Actual crop nutrient removal	38.00	4.80	53.60	208.49
Nutrient balance	-0.47	-4.80	-53.60	4,249.82
Applied to removed ratio	0.99	0.00	0.00	21.38

Fresh water applied	
13,224,000.00	*gallons*
486.99	*acre-inches*
32.47	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

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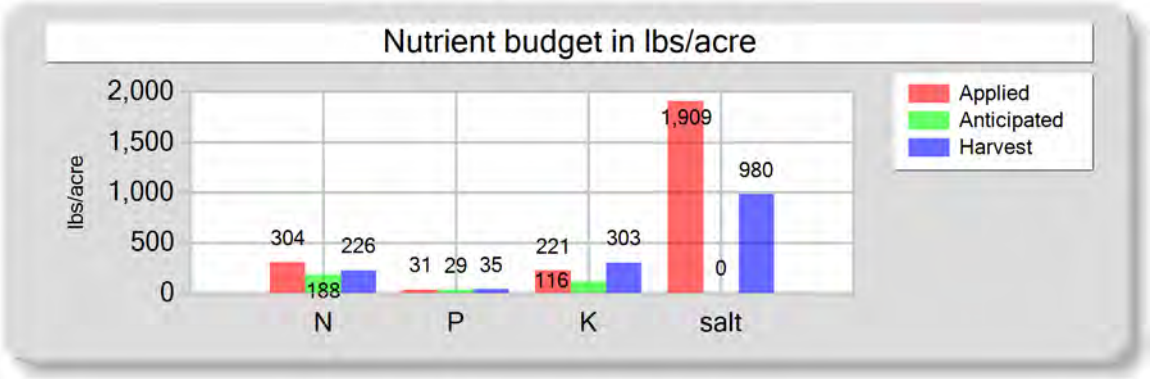
Field 3 - 10/21/2022: Wheat, silage, soft dough

Field name: Field 3 Crop: Wheat, silage, soft dough Plant date: 10/21/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	46.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	248.79	31.39	220.62	1,607.72
Fresh water	2.67	0.00	0.00	301.25
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	304.46	31.39	220.62	1,908.96
Anticipated crop nutrient removal	187.60	29.40	116.20	0.00
Actual crop nutrient removal	225.68	35.46	303.05	980.09
Nutrient balance	78.78	-4.07	-82.43	928.88
Applied to removed ratio	1.35	0.89	0.73	1.95

Fresh water applied	
15,402,240.00	*gallons*
567.21	*acre-inches*
8.86	*inches/acre*

Process wastewater applied	
2,250,000.00	*gallons*
82.86	*acre-inches*
1.29	*inches/acre*

Total harvests for the crop	
1	*harvests*

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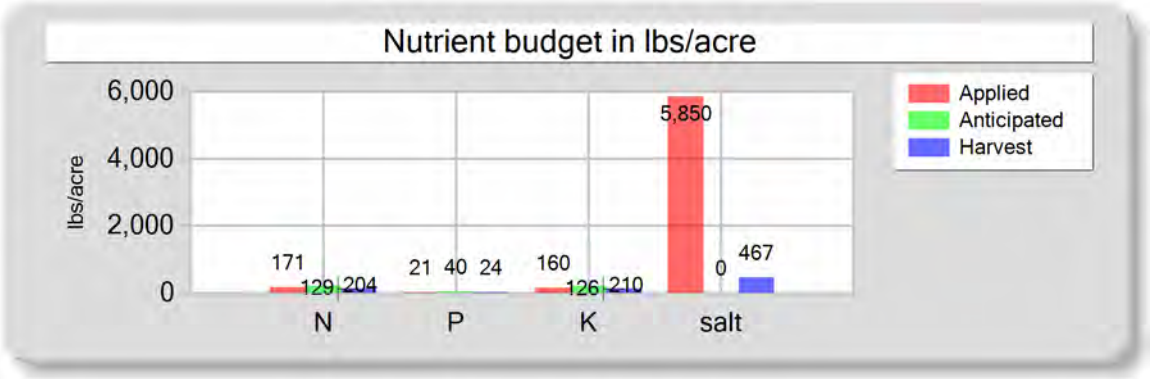
Field 3 - 06/16/2023: Corn, silage

Field name: Field 3 Crop: Corn, silage Plant date: 06/16/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	46.71	10.86	1.92	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	85.62	9.97	158.28	1,010.16
Fresh water	31.50	0.00	0.00	4,840.21
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	170.83	20.83	160.20	5,850.37
Anticipated crop nutrient removal	204.00	40.00	210.00	0.00
Actual crop nutrient removal	129.48	23.85	126.08	466.55
Nutrient balance	41.35	-3.02	34.13	5,383.82
Applied to removed ratio	1.32	0.87	1.27	12.54

Fresh water applied	
69,473,520.00	*gallons*
2,558.47	*acre-inches*
39.98	*inches/acre*

Process wastewater applied	
1,440,000.00	*gallons*
53.03	*acre-inches*
0.83	*inches/acre*

Total harvests for the crop	
1	*harvests*

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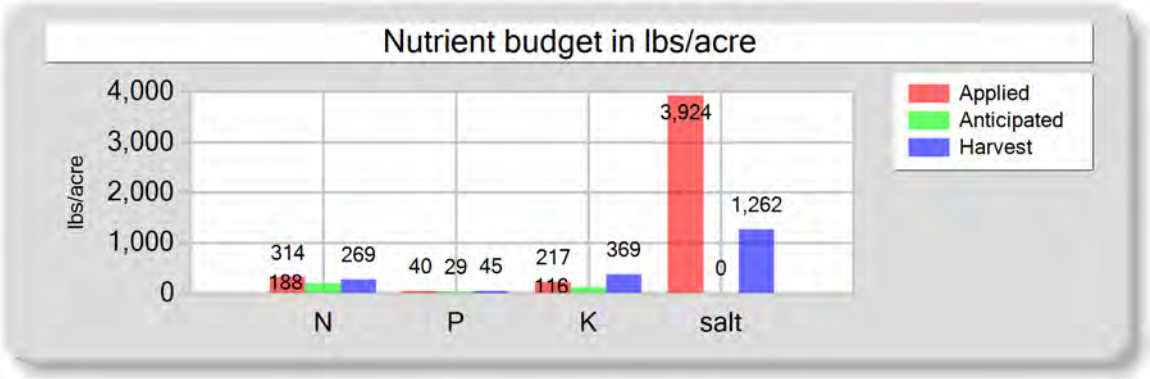
Field 6 - 10/18/2022: Wheat, silage, soft dough

Field name: Field 6 Crop: Wheat, silage, soft dough Plant date: 10/18/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	303.24	40.01	217.43	3,448.26
Fresh water	4.21	0.00	0.00	475.26
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	314.45	40.01	217.43	3,923.52
Anticipated crop nutrient removal	187.60	29.40	116.20	0.00
Actual crop nutrient removal	268.66	44.78	369.40	1,262.09
Nutrient balance	45.79	-4.77	-151.97	2,661.44
Applied to removed ratio	1.17	0.89	0.59	3.11

Fresh water applied	
18,224,640.00	*gallons*
671.15	*acre-inches*
13.98	*inches/acre*

Process wastewater applied	
2,295,000.00	*gallons*
84.52	*acre-inches*
1.76	*inches/acre*

Total harvests for the crop	
1	*harvests*

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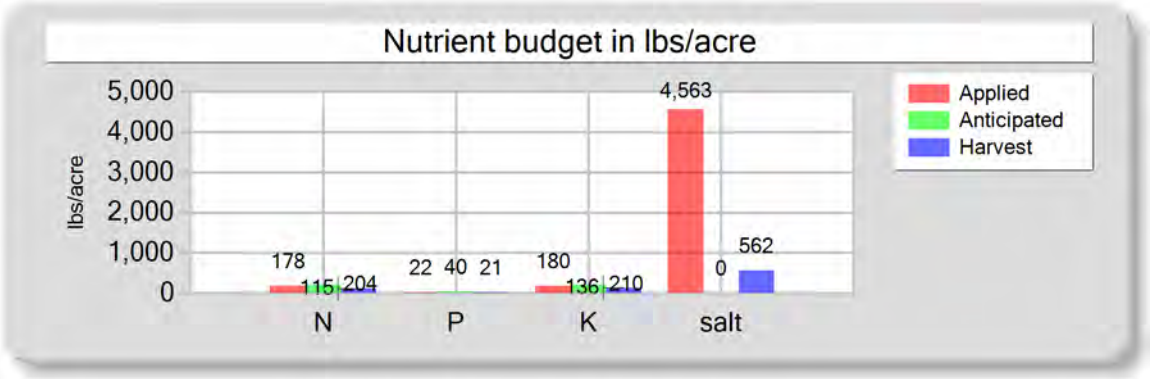
Field 6 - 06/12/2023: Corn, silage

Field name: Field 6 Crop: Corn, silage Plant date: 06/12/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	46.71	10.86	1.92	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	96.32	11.22	178.07	1,136.43
Fresh water	27.70	0.00	0.00	3,426.26
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	177.73	22.08	179.99	4,562.69
Anticipated crop nutrient removal	204.00	40.00	210.00	0.00
Actual crop nutrient removal	114.90	20.62	135.52	562.50
Nutrient balance	62.83	1.45	44.47	4,000.20
Applied to removed ratio	1.55	1.07	1.33	8.11

Fresh water applied	
51,089,040.00	*gallons*
1,881.44	*acre-inches*
39.20	*inches/acre*

Process wastewater applied	
1,215,000.00	*gallons*
44.74	*acre-inches*
0.93	*inches/acre*

Total harvests for the crop	
1	*harvests*

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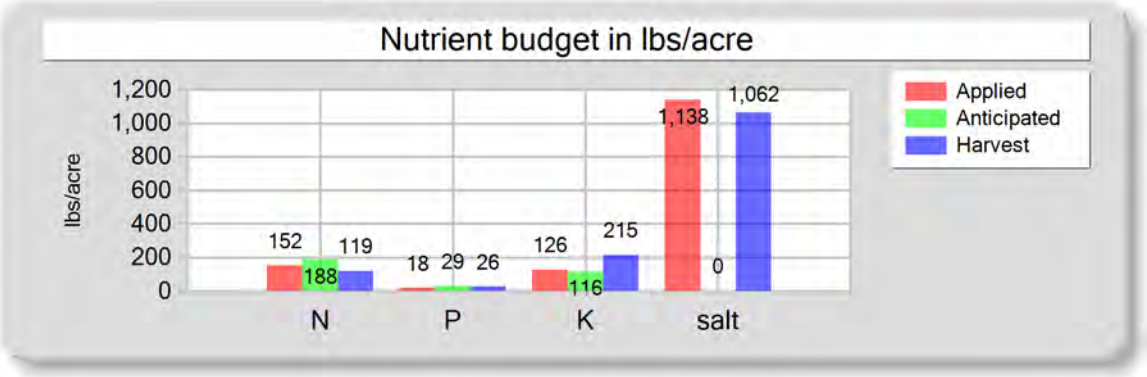
Field 7 - 10/19/2022: Wheat, silage, soft dough

Field name: Field 7 Crop: Wheat, silage, soft dough Plant date: 10/19/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	142.59	17.99	126.45	921.44
Fresh water	1.92	0.00	0.00	216.95
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	151.51	17.99	126.45	1,138.39
Anticipated crop nutrient removal	187.60	29.40	116.20	0.00
Actual crop nutrient removal	118.74	25.67	215.01	1,062.48
Nutrient balance	32.77	-7.68	-88.57	75.91
Applied to removed ratio	1.28	0.70	0.59	1.07

Fresh water applied	
11,612,160.00	*gallons*
427.64	*acre-inches*
6.38	*inches/acre*

Process wastewater applied	
1,350,000.00	*gallons*
49.72	*acre-inches*
0.74	*inches/acre*

Total harvests for the crop	
1	*harvests*

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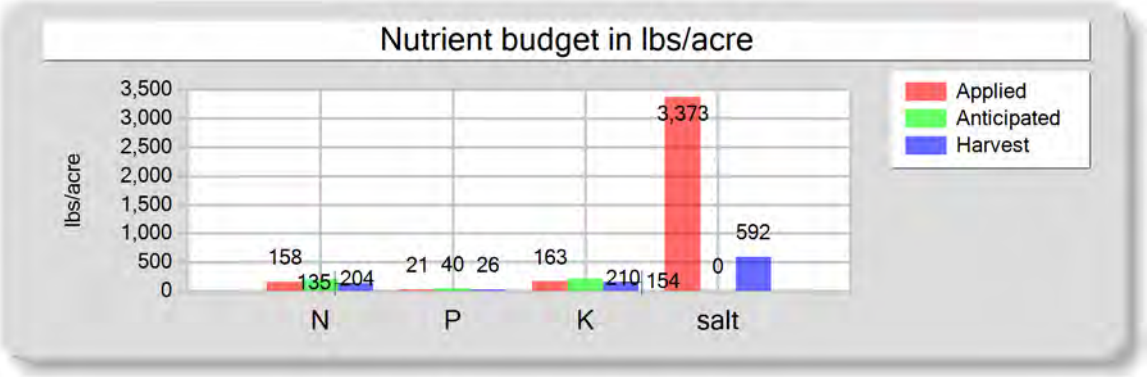
Field 7 - 06/14/2023: Corn, silage

Field name: Field 7 Crop: Corn, silage Plant date: 06/14/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	46.71	10.86	1.92	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	86.90	10.12	160.65	1,025.24
Fresh water	17.18	0.00	0.00	2,348.25
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	157.78	20.98	162.57	3,373.49
Anticipated crop nutrient removal	204.00	40.00	210.00	0.00
Actual crop nutrient removal	134.97	25.71	154.25	592.08
Nutrient balance	22.81	-4.73	8.31	2,781.41
Applied to removed ratio	1.17	0.82	1.05	5.70

Fresh water applied	
68,856,240.00	*gallons*
2,535.74	*acre-inches*
37.85	*inches/acre*

Process wastewater applied	
1,530,000.00	*gallons*
56.34	*acre-inches*
0.84	*inches/acre*

Total harvests for the crop	
1	*harvests*

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NUTRIENT ANALYSES

A. MANURE ANALYSES

Solid Manure

Sample and source description: Solid Manure
Sample date: 10/21/2022 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight
Moisture: 48.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	26,500.00	6,470.00	40,600.00							33.60
DL	100.00	100.00	100.00							0.01

Corral Solids

Sample and source description: Corral Solids
Sample date: 05/15/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight
Moisture: 66.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,500.00	5,900.00	30,900.00	18,100.00	11,500.00	5,270.00	6,300.00	1,340.00		67.10
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00		0.01

Separator Solids

Sample and source description: Separator Solids
Sample date: 05/15/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight
Moisture: 74.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	19,400.00	3,160.00	7,390.00							0.00
DL	100.00	100.00	100.00							0.01

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Separator Solids

Sample and source description: Separator Solids

Sample date: 10/05/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 73.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,000.00	2,420.00	4,970.00							67.30
DL	100.00	100.00	100.00							0.01

B. PROCESS WASTEWATER ANALYSES

4QWW

Sample and source description: 4QWW

Sample date: 12/16/2022 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	661.00	351.00	0.00	0.00	92.70	312.00								16,000.00	12,200
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

1QWW

Sample and source description: 1QWW

Sample date: 03/10/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	848.00	356.00	0.00	0.00	107.00	752.00								6,960.00	5,480
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

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2QWW

Sample and source description: 2QWW

Sample date: 05/30/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	484.00	336.00	0.00	0.00	37.20	828.00								7,180.00	8,140
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

3QWW

Sample and source description: 3QWW

Sample date: 10/03/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	456.00	300.00	0.00	0.00	53.10	843.00								7,320.00	5,380
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

4QWW

Sample and source description: 4QWW

Sample date: 12/20/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	543.00	370.00	0.00	0.00	32.60	916.00	164.00	133.00	331.00	3,280.00	0.00	222.00	17.90	7,240.00	8,180
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	1.00	0.10	0.50	0.50	1.00	10

C. FRESH WATER ANALYSES

CCID Canal

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

CCID Canal

CCID Canal

Sample description: CCID Canal

Sample date: 08/22/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1.33	0.00	0.50								246.00	150
DL	1.00	0.50	0.10								10.00	10

Drain SW

Drain SW

Sample description: Drain SW

Sample date: 08/22/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	4.15	0.00	0.00								1,001.00	606
DL	1.00	0.50	0.10								10.00	10

Irrigation Wells

IW 24

Sample description: IW 24

Sample date: 08/22/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1.00	0.00	0.40								241.00	143
DL	1.00	0.50	0.10								10.00	10

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Irrigation Wells

IW 7

Sample description: IW 7

Sample date: 08/22/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	10.10	0.00	10.10								1,920.00	1,150
DL	1.00	0.50	0.10								10.00	10

IW Hunt

Sample description: IW Hunt

Sample date: 08/22/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	4.50	0.00	4.40								1,600.00	980
DL	1.00	0.50	0.10								10.00	10

D. SOIL ANALYSES

Field 1

Soil Sample

Sample and source description: Soil Sample

Sample date: 10/17/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			134.00				
DL			2.00				

Field 14

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Field 14

Soil Sample

Sample and source description: Soil Sample

Sample date: 10/17/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			59.00				
DL			2.00				

Field 18/19

Soil Sample

Sample and source description: Soil Sample

Sample date: 11/09/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			245.00				
DL			2.00				

Field 2

Soil Sample

Sample and source description: Soil Sample

Sample date: 10/17/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			123.00				
DL			2.00				

Field 20

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Field 20

Soil Sample

Sample and source description: Soil Sample

Sample date: 11/09/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			148.00				
DL			2.00				

Field 3

Soil Sample

Sample and source description: Soil Sample

Sample date: 10/17/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			113.00				
DL			2.00				

Field 6

Soil Sample

Sample and source description: Soil Sample

Sample date: 10/17/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			134.00				
DL			2.00				

Field 7

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Field 7

Soil Sample

Sample and source description: Soil Sample

Sample date: 11/09/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			102.00				
DL			2.00				

E. PLANT TISSUE ANALYSES

Field 1 - 10/20/2022: Wheat, silage, soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/18/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 69.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,000.00	900.00	7,800.00		10.70
DL	100.00	100.00	100.00		0.01

Field 1 - 06/19/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/12/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 71.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,900.00	600.00	300.00		4.60
DL	100.00	100.00	100.00		0.01

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Field 10 - 10/21/2021: Alfalfa, hay

Alfalfa

Sample and source description: Alfalfa
Sample date: 09/12/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 7.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	33,700.00	3,600.00	20,700.00		10.90
DL	100.00	100.00	100.00		0.01

Field 14 - 10/25/2022: Wheat, silage, soft dough

Winter Forage

Sample and source description: Winter Forage
Sample date: 05/18/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 70.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,400.00	1,100.00	9,300.00		10.90
DL	100.00	100.00	100.00		0.01

Field 14 - 06/20/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage
Sample date: 10/12/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 67.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,300.00	700.00	4,100.00		4.10
DL	100.00	100.00	100.00		0.01

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Field 18/19 - 10/03/2022: Rye Grass Silage

Rye Grass Silage

Sample and source description: Rye Grass Silage
Sample date: 06/02/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 56.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,500.00	1,600.00	7,400.00		11.15
DL	100.00	100.00	100.00		0.01

Field 18/19 - 07/26/2023: Milo Silage

Milo Silage

Sample and source description: Milo Silage
Sample date: 11/07/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 48.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,500.00	1,200.00	13,400.00		10.20
DL	100.00	100.00	100.00		0.01

Field 2 - 10/27/2022: Wheat, silage, soft dough

Winter Forage

Sample and source description: Winter Forage
Sample date: 05/18/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 66.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,900.00	700.00	6,700.00		9.00
DL	100.00	100.00	100.00		0.01

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Field 2 - 06/17/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage
Sample date: 10/12/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 72.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,200.00	600.00	4,000.00		5.40
DL	100.00	100.00	100.00		0.01

Field 20 - 10/03/2022: Rye Grass Silage

Rye Grass Silage

Sample and source description: Rye Grass Silage
Sample date: 06/12/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 64.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,800.00	1,000.00	9,100.00		12.55
DL	100.00	100.00	100.00		0.01

Field 20 - 07/28/2023: Milo Silage

Milo Silage

Sample and source description: Milo Silage
Sample date: 11/07/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 58.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,200.00	800.00	8,500.00		8.80
DL	100.00	100.00	100.00		0.01

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Field 21 - 11/01/2002: Pasture

Pasture

Sample and source description: Pasture

Sample date: 11/14/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 66.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,700.00	1,000.00	9,700.00		12.40
DL	100.00	100.00	100.00		0.01

Field 21 - 10/26/2022: Wheat, silage, soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/18/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 60.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,500.00	900.00	8,100.00		9.80
DL	100.00	100.00	100.00		0.01

Field 22 - 11/01/2002: Pasture

Pasture

Sample and source description: Pasture

Sample date: 11/13/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 63.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	8,300.00	1,100.00	10,000.00		9.83
DL	100.00	100.00	100.00		0.01

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Field 23 - 10/01/2022: Rye Grass Silage

Rye Grass Silage

Sample and source description: Rye Grass Silage
Sample date: 06/12/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 58.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	8,500.00	2,900.00	2,800.00		15.90
DL	100.00	100.00	100.00		0.01

Field 23 - 07/21/2023: Milo Silage

Milo Silage

Sample and source description: Milo Silage
Sample date: 11/07/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 48.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,500.00	1,200.00	13,400.00		10.20
DL	100.00	100.00	100.00		0.01

Field 24 - 11/01/2022: Alfalfa, hay

Alfalfa

Sample and source description: Alfalfa
Sample date: 09/14/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 10.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	35,700.00	3,100.00	19,000.00		11.20
DL	100.00	100.00	100.00		0.01

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Field 25 - 10/21/2021: Alfalfa, hay

Alfalfa

Sample and source description: Alfalfa

Sample date: 09/13/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 9.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	29,200.00	3,400.00	19,400.00		10.95
DL	100.00	100.00	100.00		0.01

Field 26 - 11/03/2022: Alfalfa, hay

Alfalfa

Sample and source description: Alfalfa

Sample date: 09/14/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 20.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	29,000.00	2,500.00	16,800.00		11.35
DL	100.00	100.00	100.00		0.01

Field 27 - 11/01/2002: Pasture

Pasture

Sample and source description: Pasture

Sample date: 11/11/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 64.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,900.00	1,000.00	9,900.00		10.30
DL	100.00	100.00	100.00		0.01

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Field 28&29 - 10/01/2022: Rye Grass Silage

Rye Grass Silage

Sample and source description: Rye Grass Silage
Sample date: 06/12/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 56.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	8,100.00	800.00	2,800.00		18.20
DL	100.00	100.00	100.00		0.01

Field 28&29 - 07/25/2023: Milo Silage

Milo Silage

Sample and source description: Milo Silage
Sample date: 11/07/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 48.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,500.00	1,200.00	13,400.00		10.20
DL	100.00	100.00	100.00		0.01

Field 3 - 10/21/2022: Wheat, silage, soft dough

Winter Forage

Sample and source description: Winter Forage
Sample date: 05/18/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 69.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,000.00	1,100.00	9,400.00		10.00
DL	100.00	100.00	100.00		0.01

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Field 3 - 06/16/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage
Sample date: 10/12/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 67.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,800.00	700.00	3,700.00		4.20
DL	100.00	100.00	100.00		0.01

Field 6 - 10/18/2022: Wheat, silage, soft dough

Winter Forage

Sample and source description: Winter Forage
Sample date: 05/18/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 69.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,200.00	1,200.00	9,900.00		11.20
DL	100.00	100.00	100.00		0.01

Field 6 - 06/12/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage
Sample date: 10/12/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 68.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,900.00	700.00	4,600.00		6.10
DL	100.00	100.00	100.00		0.01

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Field 7 - 10/19/2022: Wheat, silage, soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/18/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 64.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,700.00	800.00	6,700.00		9.30
DL	100.00	100.00	100.00		0.01

Field 7 - 06/14/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/12/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 67.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,200.00	800.00	4,800.00		5.60
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

No subsurface (tile) drainage analyses entered.

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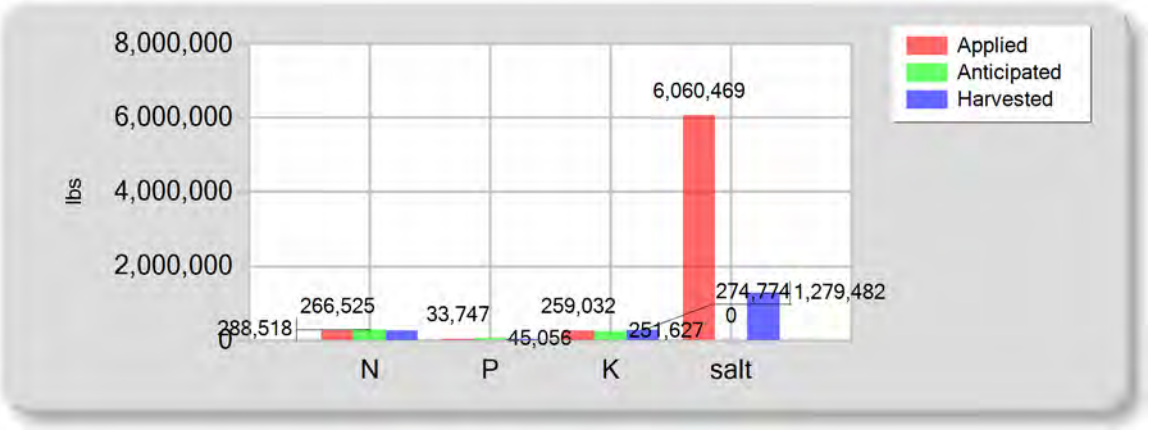
NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	22,196.44	3,953.04	698.88	0.00
Dry manure	36,128.72	8,820.86	55,351.93	458,084.93
Process wastewater	172,120.38	20,972.73	202,981.40	1,713,047.41
Fresh water	25,733.02	0.00	0.00	3,889,337.10
Atmospheric deposition	10,346.00	0.00	0.00	0.00
Total nutrients applied	266,524.56	33,746.63	259,032.21	6,060,469.44
Anticipated crop nutrient removal	288,518.00	45,055.60	251,627.00	0.00
Actual crop nutrient removal	261,625.03	40,466.27	274,773.82	1,279,481.98
Nutrient balance	4,899.53	-6,719.64	-15,741.61	4,780,987.47
Applied to removed ratio	1.02	0.83	0.94	4.74

B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL



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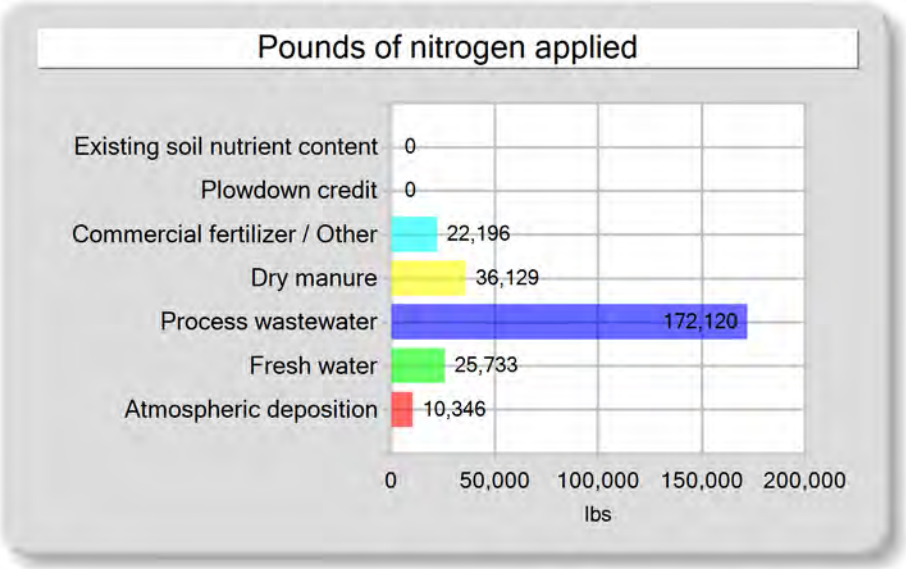
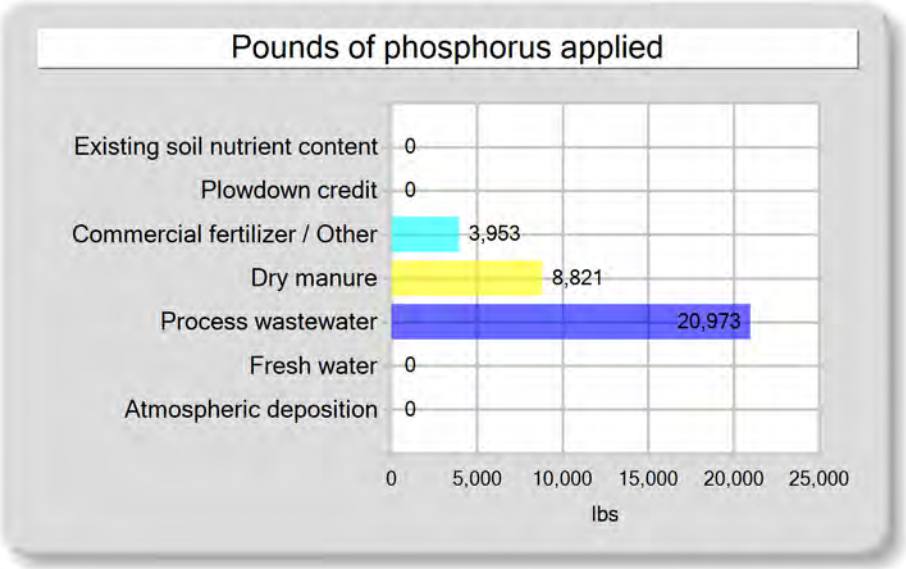
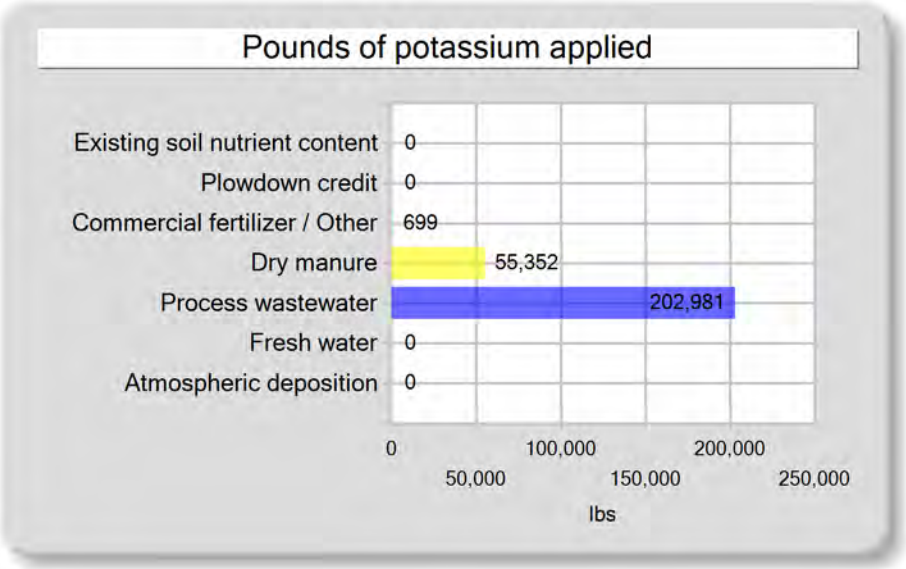
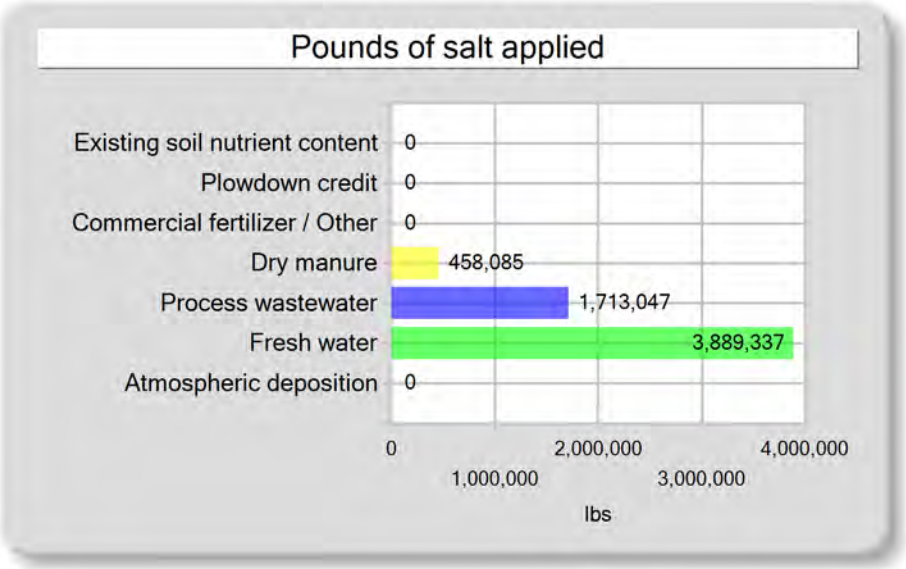
C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









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EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

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ADDITIONAL NOTES

A. NOTES

Precipitation was utilized during winter months to meet forage freshwater requirements.

All wells were samples, but heavy rains during the winter season allowed for a greater amount of surface water allocation to grow crops.

Fields 18-19 Rye Grass, 20 Rye Grass, 22 Pasture, & 27 Pasture had higher than anticipated removal rates. This was due to higher than expected %N and/or yield. The %N removed, which is based on analysis derived through a certified laboratory, exceeded the expected values for this crop. Appropriate management practices resulted in decent yields and higher than expected nutrient removal rates.

Fields 2 Winter Forage, 7 Winter Forage, 21 Winter Forage, 3 Corn, 7 Corn, 14 Corn, 18-19 Milo, 20 Milo, 23 Milo, & 28&29 Milo had lower than anticipated removal rates. This was due to much lower than expected yields and/or a lower than expected %N. The %N was based on analysis that was derived through a certified laboratory. However, the applications to these fields matched the low removal rates and was able to meet the field ratio threshold of 1.4.

Fields 2 Corn & 6 Corn had lower than anticipated removal rates due to lower than anticipated %N & low tonnage. This resulted in field ratios slightly exceeding target limits.

Fields 10 Alfalfa, 18-19 Milo, 20 Milo, 23 Milo, 25 Alfalfa, 28&29 Rye Grass, 28&29 Milo received no wastewater or solid manure in 2023. All nutrients applied to these fields were contributed through freshwater applications and/or commercial fertilizer only. Producer thought milo crops were a complete loss due to late plant dates & poor germination, so no nutrients were applied to these fields.

Field 30 was fallow for all 2023.

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

DocuSigned by:
Brad Nightengale
CE28AA3BFB3E428...

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Brad Nightengale	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
6/16/2024	
DATE	DATE

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

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Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

INSTRUCTIONS

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, complete a separate form for each destination.
3) The operator must obtain the signature of the hauler upon completion of each manure/process wastewater hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

OPERATOR INFORMATION

Name of Operator: Brad Nightengale

Name of Dairy Facility: Bar Mac Dairy

Facility Address:

Number and Street	City	County	Zip Code
5070 S Hunt RD	Gustine	Merced	95322

Contact Person Name and Phone Number: Brad Nightengale (Name) (209) 617-3087 (Phone Number)

MANURE HAULER INFORMATION

Name of Hauling Company/Person: Five Star Custom Spreading Inc.

Address of Hauling Company/Person:

Number and Street	City	State	Zip Code
3410 S Hunt DR	Gustine	CA	95322

Contact Person: Eric Rocha (Name) (209) 678-2917 (Phone Number)

DESTINATION INFORMATION

Composting Facility / Broker / Farmer / Other (identify): Composting Facility

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Five Star Custom Spreading Inc. (Name) (209) 678-2917 (Phone Number)

Address	City	State	Zip Code
3410 S Hunt DR	Gustine	CA	95322

Destination Address or Assessor's Parcel Number:

Address	City	Zip Code
3410 S Hunt RD	Gustine	95322

Street and nearest cross street (if no address) — County: Merced

Assessor's Parcel Number — Assessor's Parcel Number County

Last date hauled: 05/29/2023

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

MANURE AMOUNT HAULED

Enter the amount of manure hauled in tons, manure solids content, and the method used to calculate the amount:

Manure: 4,650.00 tons

Manure Solids Content: 34.0 %

Method used to determine amount of manure:

Number of loads multiplied by load weight

CERTIFICATION

I declare under penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true , accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

DocuSigned by:
Brad Nightengale
CE28AA3BFB3F428...
6/16/2024

Operator Signature Date

DocuSigned by:
Eric Rocha
CAEA7DE1F9EC467...
6/17/2024

Hauler Signature Date

DocuSign Envelope ID: B6960D42-6491-41E3-A4D3-F27686BB2DC0

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

INSTRUCTIONS

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2) If there are multiple destinations, complete a separate form for each destination.
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4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

OPERATOR INFORMATION

Name of Operator: Brad Nightengale

Name of Dairy Facility: Bar Mac Dairy

Facility Address:

Number and Street	City	County	Zip Code
5070 S Hunt RD	Gustine	Merced	95322

Contact Person Name and Phone Number:

Name	Phone Number
Brad Nightengale	(209) 617-3087

MANURE HAULER INFORMATION

Name of Hauling Company/Person: J&F Fertilizer

Address of Hauling Company/Person:

Number and Street	City	State	Zip Code
900 Meredith AVE	Gustine	CA	95322

Contact Person:

Name	Phone Number
John Nunes	(209) 854-6325

DESTINATION INFORMATION

Composting Facility / Broker / Farmer / Other (identify): Composting Facility

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Name	Phone Number
J&F Fertilizer	(209) 854-6325

Address	City	State	Zip Code
900 Meredith AVE	Gustine	CA	95322

Destination Address or Assessor's Parcel Number:

Address	City	Zip Code
900 Meredith AVE	Gustine	95322

Street and nearest cross street (if no address)	County
	Merced

Assessor's Parcel Number	Assessor's Parcel Number County

Last date hauled: 04/28/2023

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

MANURE AMOUNT HAULED

Enter the amount of manure hauled in tons, manure solids content, and the method used to calculate the amount:

Manure: 3,960.00 tons

Manure Solids Content: 25.3 %

Method used to determine amount of manure:

Number of loads multiplied by load weight

CERTIFICATION

I declare under penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true , accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

DocuSigned by:
Brad Nightengale
CE28AA3BFB3F428...
6/16/2024

Operator Signature | Date

DocuSigned by:
John Nunes
CAEA7DE1F9EC467...
6/17/2024

Hauler Signature | Date

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

INSTRUCTIONS

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, complete a separate form for each destination.
3) The operator must obtain the signature of the hauler upon completion of each manure/process wastewater hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

OPERATOR INFORMATION

Name of Operator: Brad Nightengale

Name of Dairy Facility: Bar Mac Dairy

Facility Address:

Number and Street	City	County	Zip Code
5070 S Hunt RD	Gustine	Merced	95322

Contact Person Name and Phone Number:

Name	Phone Number
Brad Nightengale	(209) 617-3087

MANURE HAULER INFORMATION

Name of Hauling Company/Person: J&F Fertilizer

Address of Hauling Company/Person:

Number and Street	City	State	Zip Code
900 Meredith AVE	Gustine	CA	95322

Contact Person:

Name	Phone Number
John Nunes	(209) 854-6325

DESTINATION INFORMATION

Composting Facility / Broker / Farmer / Other (identify): Composting Facility

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Name	Phone Number
J&F Fertilizer	(209) 854-6325

Address	City	State	Zip Code
900 Meredith AVE	Gustine	CA	95322

Destination Address or Assessor's Parcel Number:

Address	City	Zip Code
900 Meredith AVE	Gustine	95322

Street and nearest cross street (if no address)	County
	Merced

Assessor's Parcel Number	Assessor's Parcel Number County

Last date hauled: 12/16/2023

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

MANURE AMOUNT HAULED

Enter the amount of manure hauled in tons, manure solids content, and the method used to calculate the amount:

Manure: 6,600.00 tons

Manure Solids Content: 26.4 %

Method used to determine amount of manure:

Number of loads multiplied by load weight

CERTIFICATION

I declare under penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true , accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

DocuSigned by:
Brad Mightengale
CE28AA3BFB3F428...
Operator Signature — Date: 6/16/2024

DocuSigned by:
John Nunes
CAEA7DE1F9EC467...
Hauler Signature — Date: 6/17/2024

DocuSign Envelope ID: B6960D42-6491-41E3-A4D3-F27686BB2DC0

DELLAVALLE™
LABORATORY INC

Bar Mac Dairy/5N Cattle
26711 Taglio Road
Gustine, CA 95322

Account# 00-0023620
Account Manager: Ben Nydam
Submitted By: FR
Ranch: BM

Received: 08/22/2023 13:15
Reported: 08/29/2023 15:23

Samples in this Report

Lab ID	Sample	Matrix	Sampled By	Crop	Date Sampled
23H1909-01	CCID Canal	Canal Water	DJF		08/22/2023 0:00
23H1909-02	Tile Drain	Canal Water	DJF		08/22/2023 0:00
23H1909-03	DW - Hunt	Canal Water	DJF		08/22/2023 0:00
23H1909-04	DW - 7	Canal Water	DJF		08/22/2023 0:00
23H1909-05	DW - 24	Canal Water	DJF		08/22/2023 0:00

Default Cooler Temperature on Receipt °C: 1.4

Containers Intact
COC/Labels Agree
Preservation Confirmed
Received On Ice

Notes and Definitions

Item	Definition
H	Hold Time Exceeded
MCL	Drinking Water Maximum Contaminant Level
ND	Analyte NOT DETECTED at or above the reporting limit.
NES	Not Enough Sample
*	Not Taken
RPD	Relative Percent Difference
%REC	Percent Recovery
Source	Sample that was matrix spiked or duplicated.

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

1910 W. McKinley Ave Suite 110 Fresno, CA 93728 559-233-6129 www.dellavallelab.com

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DELLAVALLE™ LABORATORY INC

Bar Mac Dairy/5N Cattle
26711 Taglio Road
Gustine, CA 95322

Account# 00-0023620
Account Manager: Ben Nydam
Submitted By: FR
Ranch: BM

Received: 08/22/2023 13:15
Reported: 08/29/2023 15:23

Sample Results

Sample: CCID Canal
23H1909-01 (Water)

Sampled: 8/22/2023 0:00
Sampled By: DJF

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.25**	mmhos/cm	0.01	1		08/25/23 14:07	SM 2510 B		BEH1108
Electrical Conductivity umhos	**246**	umhos/cm	10.0	1		08/25/23 14:07	SM 2510 B		BEH1108
Nitrate Nitrogen as NO3N	**0.5**	mg/L	0.1	1	10	08/22/23 23:17	EPA 300.0		BEH1061
pH	**7.7**	units	1.0	1		08/25/23 14:07	SM 4500-H+	H	BEH1108
Total Filterable Solids (TDS)	**150**	mg/L	10.0	1		08/29/23 13:15	SM 2540 C		BEH1110
Temperature	**25.0**	°C	0.0	1		08/25/23 14:07	SM 2510 B		BEH1108
Kjeldahl Nitrogen (TKN), Total	ND	mg/L	1.00	1		08/28/23 13:44	SM 4500-NH3 C		BEH1183
Total Nitrogen	**1.33**	mg/L	1.00	1		08/28/23 13:44	SM 4500-NH3 C		BEH1183

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DELLAVALLE™ LABORATORY INC

Bar Mac Dairy/5N Cattle
26711 Taglio Road
Gustine, CA 95322

Account# 00-0023620
Account Manager: Ben Nydam
Submitted By: FR
Ranch: BM

Received: 08/22/2023 13:15
Reported: 08/29/2023 15:23

Sample Results
(Continued)

Sample: Tile Drain
23H1909-02 (Water)

Sampled: 8/22/2023 0:00
Sampled By: DJF

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**3.80**	mmhos/cm	0.01	1		08/25/23 14:08	SM 2510 B		BEH1108
Electrical Conductivity umhos	**3800**	umhos/cm	10.0	1		08/25/23 14:08	SM 2510 B		BEH1108
Nitrate Nitrogen as NO3N	**16.9**	mg/L	0.1	1	10	08/22/23 23:37	EPA 300.0		BEH1061
pH	**7.3**	units	1.0	1		08/25/23 14:08	SM 4500-H+	H	BEH1108
Total Filterable Solids (TDS)	**2380**	mg/L	10.0	1		08/29/23 13:15	SM 2540 C		BEH1110
Temperature	**25.0**	°C	0.0	1		08/25/23 14:08	SM 2510 B		BEH1108
Kjeldahl Nitrogen (TKN), Total	**1.44**	mg/L	1.00	1		08/28/23 13:45	SM 4500-NH3 C		BEH1183
Total Nitrogen	**18.3**	mg/L	1.00	1		08/28/23 13:45	SM 4500-NH3 C		BEH1183

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DELLAVALLE™ LABORATORY INC

Bar Mac Dairy/5N Cattle
26711 Taglio Road
Gustine, CA 95322

Account# 00-0023620
Account Manager: Ben Nydam
Submitted By: FR
Ranch: BM

Received: 08/22/2023 13:15
Reported: 08/29/2023 15:23

Sample Results
(Continued)

Sample: DW - Hunt
23H1909-03 (Water)

Sampled: 8/22/2023 0:00
Sampled By: DJF

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.60**	mmhos/cm	0.01	1		08/25/23 14:10	SM 2510 B		BEH1108
Electrical Conductivity umhos	**1600**	umhos/cm	10.0	1		08/25/23 14:10	SM 2510 B		BEH1108
Nitrate Nitrogen as NO3N	**4.4**	mg/L	0.1	1	10	08/22/23 23:56	EPA 300.0		BEH1061
pH	**7.9**	units	1.0	1		08/25/23 14:10	SM 4500-H+	H	BEH1108
Total Filterable Solids (TDS)	**980**	mg/L	10.0	1		08/29/23 13:15	SM 2540 C		BEH1110
Temperature	**25.0**	°C	0.0	1		08/25/23 14:10	SM 2510 B		BEH1108
Kjeldahl Nitrogen (TKN), Total	ND	mg/L	1.00	1		08/28/23 13:47	SM 4500-NH3 C		BEH1183
Total Nitrogen	**4.47**	mg/L	1.00	1		08/28/23 13:47	SM 4500-NH3 C		BEH1183

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DELLAVALLE™ LABORATORY INC

Bar Mac Dairy/5N Cattle
26711 Taglio Road
Gustine, CA 95322

Account# 00-0023620
Account Manager: Ben Nydam
Submitted By: FR
Ranch: BM

Received: 08/22/2023 13:15
Reported: 08/29/2023 15:23

Sample Results
(Continued)

Sample: DW - 7
23H1909-04 (Water)

Sampled: 8/22/2023 0:00
Sampled By: DJF

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.92**	mmhos/cm	0.01	1		08/25/23 14:11	SM 2510 B		BEH1108
Electrical Conductivity umhos	**1920**	umhos/cm	10.0	1		08/25/23 14:11	SM 2510 B		BEH1108
Nitrate Nitrogen as NO3N	**10.1**	mg/L	0.1	1	10	08/23/23 00:16	EPA 300.0		BEH1061
pH	**7.6**	units	1.0	1		08/25/23 14:11	SM 4500-H+	H	BEH1108
Total Filterable Solids (TDS)	**1150**	mg/L	10.0	1		08/29/23 13:15	SM 2540 C		BEH1110
Temperature	**25.0**	°C	0.0	1		08/25/23 14:11	SM 2510 B		BEH1108
Kjeldahl Nitrogen (TKN), Total	ND	mg/L	1.00	1		08/28/23 13:48	SM 4500-NH3 C		BEH1183
Total Nitrogen	**10.1**	mg/L	1.00	1		08/28/23 13:48	SM 4500-NH3 C		BEH1183

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DELLAVALLE™ LABORATORY INC

Bar Mac Dairy/5N Cattle
26711 Taglio Road
Gustine, CA 95322

Account# 00-0023620
Account Manager: Ben Nydam
Submitted By: FR
Ranch: BM

Received: 08/22/2023 13:15
Reported: 08/29/2023 15:23

Sample Results
(Continued)

Sample: DW - 24
23H1909-05 (Water)

Sampled: 8/22/2023 0:00
Sampled By: DJF

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.24**	mmhos/cm	0.01	1		08/25/23 14:13	SM 2510 B		BEH1108
Electrical Conductivity umhos	**241**	umhos/cm	10.0	1		08/25/23 14:13	SM 2510 B		BEH1108
Nitrate Nitrogen as NO3N	**0.4**	mg/L	0.1	1	10	08/23/23 00:36	EPA 300.0		BEH1061
pH	**8.2**	units	1.0	1		08/25/23 14:13	SM 4500-H+	H	BEH1108
Total Filterable Solids (TDS)	**143**	mg/L	10.0	1		08/29/23 13:15	SM 2540 C		BEH1110
Temperature	**25.0**	°C	0.0	1		08/25/23 14:13	SM 2510 B		BEH1108
Kjeldahl Nitrogen (TKN), Total	ND	mg/L	1.00	1		08/28/23 13:49	SM 4500-NH3 C		BEH1183
Total Nitrogen	ND	mg/L	1.00	1		08/28/23 13:49	SM 4500-NH3 C		BEH1183

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DELLAVALLE™ LABORATORY INC

Bar Mac Dairy/5N Cattle
26711 Taglio Road
Gustine, CA 95322

Account# 00-0023620
Account Manager: Ben Nydam
Submitted By: FR
Ranch: BM

Received: 08/22/2023 13:15
Reported: 08/29/2023 15:23

Quality Control

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEH1061										
Blank (BEH1061-BLK1)					Prepared & Analyzed: 8/22/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEH1061-BLK2)					Prepared & Analyzed: 8/22/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEH1061-BLK3)					Prepared: 8/22/2023 Analyzed: 8/23/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
LCS (BEH1061-BS1)					Prepared & Analyzed: 8/22/2023					
Nitrate Nitrogen as NO3N	5.1		0.1	mg/L	5.000		102	90-110		
LCS (BEH1061-BS2)					Prepared: 8/22/2023 Analyzed: 8/23/2023					
Nitrate Nitrogen as NO3N	5.0		0.1	mg/L	5.000		100	90-110		
Duplicate (BEH1061-DUP1)		**Source: 23H1832-01**			Prepared & Analyzed: 8/22/2023					
Nitrate Nitrogen as NO3N	0.03		0.1	mg/L		0.03			0.00	10
Duplicate (BEH1061-DUP2)		**Source: 23H1909-01**			Prepared: 8/22/2023 Analyzed: 8/23/2023					
Nitrate Nitrogen as NO3N	0.5		0.1	mg/L		0.5			0.218	10
Matrix Spike (BEH1061-MS1)		**Source: 23H1832-01**			Prepared & Analyzed: 8/22/2023					
Nitrate Nitrogen as NO3N	5.1		0.1	mg/L	5.000	0.03	102	90-110		
Matrix Spike (BEH1061-MS2)		**Source: 23H1909-01**			Prepared: 8/22/2023 Analyzed: 8/23/2023					
Nitrate Nitrogen as NO3N	5.6		0.1	mg/L	5.000	0.5	102	90-110		
Reference (BEH1061-SRM1)					Prepared & Analyzed: 8/22/2023					
Nitrate Nitrogen as NO3N	9.9			mg/L	10.00		99.1	90-110		
Reference (BEH1061-SRM2)					Prepared & Analyzed: 8/22/2023					
Nitrate Nitrogen as NO3N	9.9			mg/L	10.00		99.2	90-110		
Reference (BEH1061-SRM3)					Prepared: 8/22/2023 Analyzed: 8/23/2023					
Nitrate Nitrogen as NO3N	9.9			mg/L	10.00		99.2	90-110		

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DELLAVALLE™ LABORATORY INC

Bar Mac Dairy/5N Cattle
26711 Taglio Road
Gustine, CA 95322

Account# 00-0023620
Account Manager: Ben Nydam
Submitted By: FR
Ranch: BM

Received: 08/22/2023 13:15
Reported: 08/29/2023 15:23

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEH1108										
Blank (BEH1108-BLK1)					Prepared: 8/22/2023	Analyzed: 8/25/2023				
Electrical Conductivity	ND		0.01	mmhos/cm						
pH	5.9		1.0	units						
Temperature	25.0		0.0	°C						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Blank (BEH1108-BLK2)					Prepared: 8/22/2023	Analyzed: 8/25/2023				
pH	6.1		1.0	units						
Electrical Conductivity	ND		0.01	mmhos/cm						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Temperature	25.0		0.0	°C						
Blank (BEH1108-BLK3)					Prepared: 8/22/2023	Analyzed: 8/25/2023				
pH	6.4		1.0	units						
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	°C						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Duplicate (BEH1108-DUP1)		**Source: 23H1909-05**			Prepared: 8/22/2023	Analyzed: 8/25/2023				
pH	8.2		1.0	units		8.2			0.731	10
Electrical Conductivity	0.24		0.01	mmhos/cm		0.24			1.34	10
Electrical Conductivity umhos	237		10.0	umhos/cm		241			1.34	10
Duplicate (BEH1108-DUP2)		**Source: 23H1938-02**			Prepared: 8/22/2023	Analyzed: 8/25/2023				
pH	7.4		1.0	units		7.4			0.00	10
Electrical Conductivity	1.90		0.01	mmhos/cm		1.90			0.247	10
Electrical Conductivity umhos	1900		10.0	umhos/cm		1900			0.247	10
Reference (BEH1108-SRM1)					Prepared: 8/22/2023	Analyzed: 8/25/2023				
Electrical Conductivity	522			umhos/cm	538.0		97.0	90-110		
Reference (BEH1108-SRM2)					Prepared: 8/22/2023	Analyzed: 8/25/2023				
pH	5.8			units	5.820		100	28178-101.71		
Reference (BEH1108-SRM3)					Prepared: 8/22/2023	Analyzed: 8/25/2023				
Electrical Conductivity	962			umhos/cm	1000		96.2	90-110		
Electrical Conductivity umhos	962			umhos/cm	1000		96.2	90-110		
Reference (BEH1108-SRM4)					Prepared: 8/22/2023	Analyzed: 8/25/2023				
Electrical Conductivity	969			umhos/cm	1000		96.9	90-110		
Electrical Conductivity umhos	969			umhos/cm	1000		96.9	90-110		
Reference (BEH1108-SRM5)					Prepared: 8/22/2023	Analyzed: 8/25/2023				
Electrical Conductivity	985			umhos/cm	1000		98.5	90-110		

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DELLAVALLE™ LABORATORY INC

Bar Mac Dairy/5N Cattle
26711 Taglio Road
Gustine, CA 95322

Account# 00-0023620
Account Manager: Ben Nydam
Submitted By: FR
Ranch: BM

Received: 08/22/2023 13:15
Reported: 08/29/2023 15:23

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEH1108 (Continued)										
Reference (BEH1108-SRM5)					Prepared: 8/22/2023	Analyzed: 8/25/2023				
Electrical Conductivity umhos	985			umhos/cm	1000		98.5	90-110		
Reference (BEH1108-SRM6)					Prepared: 8/22/2023	Analyzed: 8/25/2023				
pH	4.0			units	4.000		100	97.5-102.5		
Reference (BEH1108-SRM7)					Prepared: 8/22/2023	Analyzed: 8/25/2023				
pH	4.0			units	4.000		101	97.5-102.5		
Reference (BEH1108-SRM8)					Prepared: 8/22/2023	Analyzed: 8/25/2023				
pH	4.0			units	4.000		100	97.5-102.5		

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DELLAVALLE™ LABORATORY INC

Bar Mac Dairy/5N Cattle
26711 Taglio Road
Gustine, CA 95322

Account# 00-0023620
Account Manager: Ben Nydam
Submitted By: FR
Ranch: BM

Received: 08/22/2023 13:15
Reported: 08/29/2023 15:23

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEH1110										
Blank (BEH1110-BLK1)					Prepared: 8/23/2023	Analyzed: 8/29/2023				
Total Filterable Solids (TDS)	ND		10.0	mg/L						
LCS (BEH1110-BS1)					Prepared: 8/23/2023	Analyzed: 8/29/2023				
Total Filterable Solids (TDS)	21.2		10.0	mg/L	2000		1.06	0-200		
Duplicate (BEH1110-DUP1)		**Source: 23H1909-01**			Prepared: 8/23/2023	Analyzed: 8/29/2023				
Total Filterable Solids (TDS)	140		10.0	mg/L		150			6.90	10
Duplicate (BEH1110-DUP2)		**Source: 23H1942-01**			Prepared: 8/23/2023	Analyzed: 8/29/2023				
Total Filterable Solids (TDS)	2520		10.0	mg/L		2480			1.33	10
Reference (BEH1110-SRM1)					Prepared: 8/23/2023	Analyzed: 8/29/2023				
Total Filterable Solids (TDS)	303			mg/L	325.0		93.3	90-110		

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DELLAVALLE™ LABORATORY INC

Bar Mac Dairy/5N Cattle
26711 Taglio Road
Gustine, CA 95322

Account# 00-0023620
Account Manager: Ben Nydam
Submitted By: FR
Ranch: BM

Received: 08/22/2023 13:15
Reported: 08/29/2023 15:23

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEH1183										
Blank (BEH1183-BLK1)					Prepared: 8/24/2023	Analyzed: 8/28/2023				
Kjeldahl Nitrogen (TKN), Total	ND		1.00	mg/L						
Total Nitrogen	ND		1.00	mg/L						
Blank (BEH1183-BLK2)					Prepared: 8/24/2023	Analyzed: 8/28/2023				
Kjeldahl Nitrogen (TKN), Total	ND		1.00	mg/L						
Total Nitrogen	ND		1.00	mg/L						
LCS (BEH1183-BS1)					Prepared: 8/24/2023	Analyzed: 8/28/2023				
Kjeldahl Nitrogen (TKN), Total	6.11		1.00	mg/L	5.709		107	90-110		
LCS (BEH1183-BS2)					Prepared: 8/24/2023	Analyzed: 8/28/2023				
Kjeldahl Nitrogen (TKN), Total	5.92		1.00	mg/L	5.709		104	90-110		
Duplicate (BEH1183-DUP1)		**Source: 23H1909-04**			Prepared: 8/24/2023	Analyzed: 8/28/2023				
Kjeldahl Nitrogen (TKN), Total	ND		1.40	mg/L		ND				10
Duplicate (BEH1183-DUP2)		**Source: 23H1938-01**			Prepared: 8/24/2023	Analyzed: 8/28/2023				
Kjeldahl Nitrogen (TKN), Total	ND		1.40	mg/L		ND				10
Matrix Spike (BEH1183-MS1)		**Source: 23H1909-04**			Prepared: 8/24/2023	Analyzed: 8/28/2023				
Kjeldahl Nitrogen (TKN), Total	6.19		1.40	mg/L	7.992	ND	77.5	90-110		
Matrix Spike (BEH1183-MS2)		**Source: 23H1938-01**			Prepared: 8/24/2023	Analyzed: 8/28/2023				
Kjeldahl Nitrogen (TKN), Total	7.63		1.40	mg/L	7.992	ND	95.5	90-110		
Reference (BEH1183-SRM1)					Prepared: 8/24/2023	Analyzed: 8/28/2023				
Kjeldahl Nitrogen (TKN), Total	24.4			mg/L	23.80		103	90-110		

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08/22/23 13:15 23H1909 JG

DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 · 800 228-9896 · Fax 559 268-8174

No. Samples: 5 Material: Water

Purchase Order No Bill To: Acct # 23620 | 08 Cons # 8/22/23

Name: Bar Mac Dairy/5N Cattle
Address: 5070 S. Hunt Rd.
City: Gustine **State:** CA **Zip:** 95322
Telephone: **Fax:**
Cell/Email: dejagerfarms@gmail.com krhodes@fragservices.com
COPY TO: ariordan@fragservices.com dustin@dejagerfarms.com
REQUESTED BY: FR
RANCH: Bm
ID CROP:
Present Stage of Growth
Intended Previous
Date Sampled: 8/22 **Sampled By:** DJF

Analysis Required:

( ) Manure: DM1 (N,P,K, Moisture)
( ) Manure: DM2 (DM1 Plus Ash, Ca, Mg, Na, S, Cl)
( ) Soil Spring: DS1 (NO_3-N (0'-1')
( ) Soil Fall: DS2 (FA1 (0-1'); NO_3-N (1-2'))
( ) Harvest Tissue: CRA1 (N, P, K, Moisture, Ash)
( ) Feed: ADF/NDF
( ) Petiole/Midrib: P1
( ) Leaf: L1
(X) Other: DCW1 + TN

DESCRIPTION OF SAMPLE

1	CCID Canal	1.4
2	Tile Drain	2.7
3	DW - Hunt	2.0
4	DW - 7	2.8
5	DW - 24	3.3
6		
7		
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21		
22		
23		
24		

IR Thermometer SN: 192603727
Correction Factor: 0°C
Calibration Due: 9/26/2023
Location: Laboratory

[X] QA/QC Document [X] Copy of Chain [ ] Comments/Recommendations

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	JLF	DLV	8-22-23 12:20	8-22-23 1:05
Second				
Third				
Fourth			8/22 13:15	

I guarantee that as the client, or on behalf of client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a liquidated damage fee of 2% per month (annually 24%) or $5.00 per month whichever is greater.

If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc. (*cal*). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through *cal* under its Rules and Procedures. The parties will equally bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all mediation and arbitration costs, and in the event of arbitration, reasonable attorneys' fees of Dellavalle Laboratory.

Invoicing Information:
Sampling hrs
Miles
Consulting

Shipping
$ In
$ Out

Amt Paid	Rec By	Check #	Date

Signature
Sample received in cooler with ice (coolant)
[ ] Yes [ ] No

X:\1-CLIENT CUSTOM FIELDSHEETS\FIELDSHEETS\Dairy Fieldsheets\[F & R Ag Services (Alex).xls]Soil & Tissue rev 4.18.18

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Paste Label Here

08/22/23 13:15 23H1909

Shipping Information: Shipped In ☐ Picked-Up ☒ Walk In ☐ DLI Sampler ☐ Other ☐ _____

☐ Samples refridgerated before pick up ☐ Picked up samples placed in Ice chest

Container: Ice Chest ☒ Box ☐ None ☐ **Refrigerant:** Wet Ice ☒ Blue Ice ☐ None ☐

Samples Preserved with HNO_3 or H_2SO_4 were: ☐ Received Preserved ☒ Preserved Upon Receipt at Laboratory

Type of Container(s) Received		Sample Number 1	2	3	4	5	6	7	8	9	10
Sample Containers for Internal (DLI) Use *(Containers that go into the Lab)*											
Plastics	100 mL sterile plastic $Na_2S_2O_3$ (Green)										
Plastics	250 mL unpreserved (White) Plastic										
Plastics	250 mL HNO_3 (Red) Plastic										
Plastics	* pH Value										
Plastics	250 mL H_2SO_4 (Yellow) Plastic	1	1	1	1	1					
Plastics	* pH Value	<2	<2	<2	<2	<2					
Plastics	500 mL unpreserved (White) Plastic										
Plastics	1 L unpreserved (White) Plastic	1	1	1	1	1					
Plastics	1 L unpreserved (BOD) (Purple) Plastic										
Special	500mL unpreserved (White) Glass										
Special	PO4-P Kit										
Special	Other:										
Sample Containers for Subcontracted ("Send Out") Analyses *(Containers that go in the Subcontract ("Send Out") Refrigerator)*											
Plastics	100 mL sterile plastic $Na_2S_2O_3$ (Green)										
Plastics	250 mL unpreserved (White) Plastic										
Plastics	250 mL HNO_3 (Red) Plastic										
Plastics	250 mL H_2SO_4 (Yellow) Plastic										
Plastics	500 mL HNO_3 (Red)										
Plastics	1 L unpreserved (White) Plastic										
Plastics	1 L unpreserved (BOD) (Purple) Plastic										
Plastics	1 L HNO_3 (Red)										
VOA Vials	40 mL VOA, $Na_2S_2O_3$ + MCAA (EPA531)										
VOA Vials	40 mL VOA, $Na_2S_2O_3$ (EPA547)										
VOA Vials	40mL AG VOA unpreserved (White) (Set of 3)										
VOA Vials	40 mL AG VOA, $Na_2S_2O_3$ (Green) (Set of 3)										
VOA Vials	40mL VOA, H_3PO_4 (Set of 3)										
VOA Vials	40 mL VOA, HCl (Blue) (Set of 3)										
VOA Vials	40 mL VOA, $Na_2S_2O_3$ (Green) (Set of 3)										
Glass	250 mL AG unpreserved (White)										
Glass	250 mL AG H_2SO_4 (Yellow)										
Glass	250 mL AG $Na_2S_2O_3$ (Green)										
Glass	250 mL AG $Na_2S_2O_3$ + MCAA										
Glass	500 mL glass unpreserved (White)										
Glass	500 mL AG HCl (Blue)										
Glass	1 L AG unpreserved (White)										
Glass	1 L AG H_2SO_4 (Yellow)										
Glass	1 L AG $Na_2S_2O_3$ (Green)										
Glass	1 L AG HCl (Blue)										
Special	Cr^{6+} - 50mL Plastic w/Borate/HCO_3/CO_3										
Special	Cyanide - 500 mL NaOH										
Special	Asbestos - 1L P wrapped in foil (Set of 2)										
Special	Sulfide - 1 L **AG** or **P** NaOH + ZnAc										
Special	Chlorite/Bromate - 250 mL AG with EDA										
Special	HAA5 - 250mL AG Ammonium Chlorite										
Special	DO KIT										
Special	Other:										
Special	Other:										

pH Strips
Lot 10BDH4501
Exp. Jan. 2025

SEND OUT

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DELLAVALLE™ LABORATORY INC

Bar Mac Dairy/5N Cattle
26711 Taglio Road
Gustine, CA 95322

Account# 00-0023620
Account Manager: Ben Nydam
Submitted By: Brad Nightengale
Ranch: BM / 5N

Received: 06/30/2022 13:43
Reported: 07/08/2022 11:28

Samples in this Report

Lab ID	Sample	Matrix	Sampled By	Crop	Date Sampled
22F2287-01	BM - Dom Milk Barn	Ground Water	DJF		06/29/2022 0:00
22F2287-02	BM - Dom Dairy Barn	Ground Water	DJF		06/29/2022 0:00
22F2287-03	BM - Dom Hay Barn	Ground Water	DJF		06/29/2022 0:00
22F2287-04	BM - IW Hunt	Ground Water	DJF		06/29/2022 0:00
22F2287-05	BM - IW #7	Ground Water	DJF		06/29/2022 0:00
22F2287-06	BM - IW #24	Ground Water	DJF		06/29/2022 0:00
22F2287-07	BM - CCID Canal	Ground Water	DJF		06/29/2022 0:00
22F2287-08	5N - Dom 1	Ground Water	DJF		06/29/2022 0:00
22F2287-09	5N - Dom 2	Ground Water	DJF		06/29/2022 0:00
22F2287-10	5N - Diesel Well	Ground Water	DJF		06/29/2022 0:00

Default Cooler Temperature on Receipt °C: -0.2

Containers Intact
COC/Labels Agree
Preservation Confirmed
Received On Ice

Notes and Definitions

Item	Definition
H	Hold Time Exceeded
MCL	Drinking Water Maximum Contaminant Level
ND	Analyte NOT DETECTED at or above the reporting limit.
NES	Not Enough Sample
*	Not Taken
RPD	Relative Percent Difference
%REC	Percent Recovery
Source	Sample that was matrix spiked or duplicated.

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

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1910 W. McKinley Ave Suite 110 Fresno, CA 93728 559-233-6129 www.dellavallelab.com

DELLAVALLE™ LABORATORY INC

Bar Mac Dairy/5N Cattle
26711 Taglio Road
Gustine, CA 95322

Account# 00-0023620
Account Manager: Ben Nydam
Submitted By: Brad Nightengale
Ranch: BM / 5N

Received: 06/30/2022 13:43
Reported: 07/08/2022 11:28

Sample Results

Sample: BM - Dom Milk Barn
22F2287-01 (Water)

Sampled: 6/29/2022 0:00
Sampled By: DJF

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Alkalinity as CaCO3	**257**	mg/L	10.0	1		07/05/22 13:53	SM 2320 B		BDF1245
Calcium	**82.8**	mg/L	0.1	1		07/06/22 14:01	EPA 200.7		BDG0009
Chloride	**154**	mg/L	0.2	1	250	06/30/22 21:53	EPA 300.0		BDF1188
Carbonate as CaCO3	ND	mg/L	1	1		07/05/22 13:53	SM 2320 B		BDF1245
Electrical Conductivity	**1.36**	mmhos/cm	0.01	1		07/05/22 13:53	SM 2510 B		BDF1245
Electrical Conductivity umhos	**1360**	umhos/cm	10.0	1		07/05/22 13:53	SM 2510 B		BDF1245
Bicarbonate as CaCO3	**257**	mg/L	5.00	1		07/05/22 13:53	SM 2320 B		BDF1245
Potassium	**1.64**	mg/L	0.500	1		07/06/22 14:01	EPA 200.7		BDG0009
Magnesium	**64.2**	mg/L	0.1	1		07/06/22 14:01	EPA 200.7		BDG0009
Sodium	**108**	mg/L	1	1		07/06/22 14:01	EPA 200.7		BDG0009
Ammonia as NH4-N, Total	*	mg/L	0.00	1		06/29/22 00:00	Field		BDF1252
Nitrate Nitrogen as NO3N	**6.2**	mg/L	0.1	1	10	06/30/22 21:53	EPA 300.0		BDF1188
Hydroxide as CaCO3	ND	mg/L	1.00	1		07/05/22 13:53	SM 2320 B		BDF1245
pH	**7.8**	units	1.0	1		07/05/22 13:53	SM 4500-H+	H	BDF1245
Sulfate (SO4)	**178**	mg/L	0.5	1	250	06/30/22 21:53	EPA 300.0		BDF1188
Total Filterable Solids (TDS)	**817**	mg/L	10.0	1		07/07/22 10:56	SM 2540 C		BDG0005
Kjeldahl Nitrogen (TKN), Total	ND	mg/L	1.00	1		07/05/22 14:00	SM 4500-NH3 C		BDG0010
Total Nitrogen	**6.18**	mg/L	1.00	1		07/05/22 14:00	SM 4500-NH3 C		BDG0010

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DELLAVALLE™ LABORATORY INC

Bar Mac Dairy/5N Cattle
26711 Taglio Road
Gustine, CA 95322

Account# 00-0023620
Account Manager: Ben Nydam
Submitted By: Brad Nightengale
Ranch: BM / 5N

Received: 06/30/2022 13:43
Reported: 07/08/2022 11:28

Sample Results
(Continued)

Sample: BM - Dom Dairy Barn
22F2287-02 (Water)

Sampled: 6/29/2022 0:00
Sampled By: DJF

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Alkalinity as CaCO3	**257**	mg/L	10.0	1		07/05/22 13:59	SM 2320 B		BDF1245
Calcium	**78.1**	mg/L	0.1	1		07/06/22 14:02	EPA 200.7		BDG0009
Chloride	**143**	mg/L	0.2	1	250	06/30/22 22:12	EPA 300.0		BDF1188
Carbonate as CaCO3	ND	mg/L	1	1		07/05/22 13:59	SM 2320 B		BDF1245
Electrical Conductivity	**1.29**	mmhos/cm	0.01	1		07/05/22 13:59	SM 2510 B		BDF1245
Electrical Conductivity umhos	**1290**	umhos/cm	10.0	1		07/05/22 13:59	SM 2510 B		BDF1245
Bicarbonate as CaCO3	**257**	mg/L	5.00	1		07/05/22 13:59	SM 2320 B		BDF1245
Potassium	**1.48**	mg/L	0.500	1		07/06/22 14:02	EPA 200.7		BDG0009
Magnesium	**61.1**	mg/L	0.1	1		07/06/22 14:02	EPA 200.7		BDG0009
Sodium	**103**	mg/L	1	1		07/06/22 14:02	EPA 200.7		BDG0009
Ammonia as NH4-N, Total	*	mg/L	0.00	1		06/29/22 00:00	Field		BDF1252
Nitrate Nitrogen as NO3N	**5.6**	mg/L	0.1	1	10	06/30/22 22:12	EPA 300.0		BDF1188
Hydroxide as CaCO3	ND	mg/L	1.00	1		07/05/22 13:59	SM 2320 B		BDF1245
pH	**7.8**	units	1.0	1		07/05/22 13:59	SM 4500-H+	H	BDF1245
Sulfate (SO4)	**166**	mg/L	0.5	1	250	06/30/22 22:12	EPA 300.0		BDF1188
Total Filterable Solids (TDS)	**763**	mg/L	10.0	1		07/07/22 10:56	SM 2540 C		BDG0005
Kjeldahl Nitrogen (TKN), Total	ND	mg/L	1.00	1		07/05/22 14:01	SM 4500-NH3 C		BDG0010
Total Nitrogen	**5.57**	mg/L	1.00	1		07/05/22 14:01	SM 4500-NH3 C		BDG0010

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DELLAVALLE™ LABORATORY INC

Bar Mac Dairy/5N Cattle
26711 Taglio Road
Gustine, CA 95322

Account# 00-0023620
Account Manager: Ben Nydam
Submitted By: Brad Nightengale
Ranch: BM / 5N

Received: 06/30/2022 13:43
Reported: 07/08/2022 11:28

Sample Results
(Continued)

Sample: BM - Dom Hay Barn
22F2287-03 (Water)

Sampled: 6/29/2022 0:00
Sampled By: DJF

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Alkalinity as CaCO3	**246**	mg/L	10.0	1		07/05/22 14:06	SM 2320 B		BDF1245
Calcium	**67.2**	mg/L	0.1	1		07/06/22 14:03	EPA 200.7		BDG0009
Chloride	**123**	mg/L	0.2	1	250	06/30/22 22:31	EPA 300.0		BDF1188
Carbonate as CaCO3	ND	mg/L	1	1		07/05/22 14:06	SM 2320 B		BDF1245
Electrical Conductivity	**1.18**	mmhos/cm	0.01	1		07/05/22 14:06	SM 2510 B		BDF1245
Electrical Conductivity umhos	**1180**	umhos/cm	10.0	1		07/05/22 14:06	SM 2510 B		BDF1245
Bicarbonate as CaCO3	**246**	mg/L	5.00	1		07/05/22 14:06	SM 2320 B		BDF1245
Potassium	**1.65**	mg/L	0.500	1		07/06/22 14:03	EPA 200.7		BDG0009
Magnesium	**56.4**	mg/L	0.1	1		07/06/22 14:03	EPA 200.7		BDG0009
Sodium	**96**	mg/L	1	1		07/06/22 14:03	EPA 200.7		BDG0009
Ammonia as NH4-N, Total	*	mg/L	0.00	1		06/29/22 00:00	Field		BDF1252
Nitrate Nitrogen as NO3N	**5.8**	mg/L	0.1	1	10	06/30/22 22:31	EPA 300.0		BDF1188
Hydroxide as CaCO3	ND	mg/L	1.00	1		07/05/22 14:06	SM 2320 B		BDF1245
pH	**8.0**	units	1.0	1		07/05/22 14:06	SM 4500-H+	H	BDF1245
Sulfate (SO4)	**137**	mg/L	0.5	1	250	06/30/22 22:31	EPA 300.0		BDF1188
Total Filterable Solids (TDS)	**693**	mg/L	10.0	1		07/07/22 10:56	SM 2540 C		BDG0005
Kjeldahl Nitrogen (TKN), Total	ND	mg/L	1.00	1		07/05/22 14:03	SM 4500-NH3 C		BDG0010
Total Nitrogen	**5.81**	mg/L	1.00	1		07/05/22 14:03	SM 4500-NH3 C		BDG0010

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Ranch: BM / 5N

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Quality Control

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BDF1188										
Blank (BDF1188-BLK1)					Prepared & Analyzed: 6/30/2022					
Chloride	ND		0.2	mg/L						
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Sulfate (SO4)	ND		0.5	mg/L						
Blank (BDF1188-BLK2)					Prepared & Analyzed: 6/30/2022					
Chloride	ND		0.2	mg/L						
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Sulfate (SO4)	ND		0.5	mg/L						
Blank (BDF1188-BLK3)					Prepared & Analyzed: 7/1/2022					
Chloride	ND		0.2	mg/L						
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Sulfate (SO4)	ND		0.5	mg/L						
LCS (BDF1188-BS1)					Prepared & Analyzed: 6/30/2022					
Chloride	4.6		0.2	mg/L	5.000		92.3	90-110		
Nitrate Nitrogen as NO3N	4.8		0.1	mg/L	5.000		96.2	90-110		
Sulfate (SO4)	4.6		0.5	mg/L	5.000		92.1	90-110		
LCS (BDF1188-BS2)					Prepared & Analyzed: 7/1/2022					
Chloride	4.7		0.2	mg/L	5.000		94.1	90-110		
Nitrate Nitrogen as NO3N	4.9		0.1	mg/L	5.000		97.6	90-110		
Sulfate (SO4)	4.6		0.5	mg/L	5.000		92.1	90-110		
Duplicate (BDF1188-DUP1)		**Source: 22F2217-01**			Prepared & Analyzed: 6/30/2022					
Chloride	15.1		0.2	mg/L		15.3			1.46	10
Nitrate Nitrogen as NO3N	18.4		0.1	mg/L		18.6			1.09	10
Sulfate (SO4)	38.8		0.5	mg/L		39.4			1.51	10
Duplicate (BDF1188-DUP2)		**Source: 22F2287-03**			Prepared & Analyzed: 7/1/2022					
Chloride	124.4		0.2	mg/L		123.4			0.837	10
Nitrate Nitrogen as NO3N	5.9		0.1	mg/L		5.8			1.52	10
Sulfate (SO4)	138		0.5	mg/L		137			0.608	10
Matrix Spike (BDF1188-MS1)		**Source: 22F2217-01**			Prepared & Analyzed: 6/30/2022					
Chloride	20.2		0.2	mg/L	5.000	15.3	97.5	90-110		
Nitrate Nitrogen as NO3N	23.5		0.1	mg/L	5.000	18.6	97.4	90-110		
Sulfate (SO4)	43.9		0.5	mg/L	5.000	39.4	90.1	90-110		
Matrix Spike (BDF1188-MS2)		**Source: 22F2287-03**			Prepared & Analyzed: 7/1/2022					
Chloride	126.8		0.2	mg/L	5.000	123.4	67.3	90-110		
Nitrate Nitrogen as NO3N	10.5		0.1	mg/L	5.000	5.8	94.2	90-110		
Sulfate (SO4)	141		0.5	mg/L	5.000	137	70.0	90-110		

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Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BDF1188 (Continued)										
Reference (BDF1188-SRM1)					Prepared & Analyzed: 6/30/2022					
Chloride	12.5			mg/L	12.51		99.8	90-110		
Nitrate Nitrogen as NO3N	10.0			mg/L	10.01		100	90-110		
Sulfate (SO4)	9.7			mg/L	10.01		96.7	90-110		
Reference (BDF1188-SRM2)					Prepared & Analyzed: 6/30/2022					
Chloride	12.5			mg/L	12.51		99.8	90-110		
Nitrate Nitrogen as NO3N	10.0			mg/L	10.01		100	90-110		
Sulfate (SO4)	9.6			mg/L	10.01		96.1	90-110		
Reference (BDF1188-SRM3)					Prepared & Analyzed: 7/1/2022					
Chloride	12.6			mg/L	12.51		100	90-110		
Nitrate Nitrogen as NO3N	10.1			mg/L	10.01		101	90-110		
Sulfate (SO4)	9.7			mg/L	10.01		97.0	90-110		

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Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BDF1245										
Blank (BDF1245-BLK1)					Prepared & Analyzed: 7/5/2022					
Alkalinity as CaCO3	ND		10.0	mg/L						
Carbonate as CaCO3	ND		1	mg/L						
Hydroxide as CaCO3	ND		1.00	mg/L						
Electrical Conductivity	ND		0.01	mmhos/cm						
pH	5.2		1.0	units						
Bicarbonate as CaCO3	ND		5.00	mg/L						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Blank (BDF1245-BLK2)					Prepared & Analyzed: 7/5/2022					
pH	5.5		1.0	units						
Hydroxide as CaCO3	ND		1.00	mg/L						
Electrical Conductivity	ND		0.01	mmhos/cm						
Carbonate as CaCO3	ND		1	mg/L						
Alkalinity as CaCO3	ND		10.0	mg/L						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Bicarbonate as CaCO3	ND		5.00	mg/L						
Blank (BDF1245-BLK3)					Prepared & Analyzed: 7/5/2022					
Electrical Conductivity	ND		0.01	mmhos/cm						
Carbonate as CaCO3	ND		1	mg/L						
Hydroxide as CaCO3	ND		1.00	mg/L						
pH	5.5		1.0	units						
Alkalinity as CaCO3	ND		10.0	mg/L						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Bicarbonate as CaCO3	ND		5.00	mg/L						
Duplicate (BDF1245-DUP1)		**Source: 22F2287-01**			Prepared & Analyzed: 7/5/2022					
pH	7.8		1.0	units		7.8			0.384	10
Electrical Conductivity	1.35		0.01	mmhos/cm		1.36			0.844	10
Carbonate as CaCO3	ND		1	mg/L		ND				10
Hydroxide as CaCO3	ND		1.00	mg/L		ND				10
Alkalinity as CaCO3	262		10.0	mg/L		257			1.91	10
Electrical Conductivity umhos	1350		10.0	umhos/cm		1360			0.844	10
Duplicate (BDF1245-DUP2)		**Source: 22F2287-06**			Prepared & Analyzed: 7/5/2022					
Carbonate as CaCO3	ND		1	mg/L		ND				10
Alkalinity as CaCO3	239		10.0	mg/L		242			1.45	10
Electrical Conductivity	2.03		0.01	mmhos/cm		2.04			0.427	10
Hydroxide as CaCO3	ND		1.00	mg/L		ND				10
pH	7.9		1.0	units		7.8			1.14	10
Electrical Conductivity umhos	2030		10.0	umhos/cm		2040			0.427	10
Reference (BDF1245-SRM1)					Prepared & Analyzed: 7/5/2022					

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Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BDF1245 (Continued)										
Reference (BDF1245-SRM1)					Prepared & Analyzed: 7/5/2022					
Alkalinity as CaCO3	38.1			mg/L	40.60		93.8	90-110		
Electrical Conductivity	554			umhos/cm	538.0		103	90-110		
Reference (BDF1245-SRM2)					Prepared & Analyzed: 7/5/2022					
Electrical Conductivity	545			umhos/cm	538.0		101	90-110		
Alkalinity as CaCO3	39.2			mg/L	40.60		96.6	90-110		
Reference (BDF1245-SRM3)					Prepared & Analyzed: 7/5/2022					
Electrical Conductivity	548			umhos/cm	538.0		102	90-110		
Alkalinity as CaCO3	39.5			mg/L	40.60		97.2	90-110		
Reference (BDF1245-SRM4)					Prepared & Analyzed: 7/5/2022					
pH	4.0			units	4.000		100	97.5-102.5		
Reference (BDF1245-SRM5)					Prepared & Analyzed: 7/5/2022					
pH	4.0			units	4.000		101	97.5-102.5		
Reference (BDF1245-SRM6)					Prepared & Analyzed: 7/5/2022					
pH	4.0			units	4.000		100	97.5-102.5		
Reference (BDF1245-SRM7)					Prepared & Analyzed: 7/5/2022					
pH	8.0			units	7.900		101	73418-101.26		

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Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BDG0005										
Blank (BDG0005-BLK1)					Prepared: 7/1/2022 Analyzed: 7/7/2022					
Total Filterable Solids (TDS)	ND		10.0	mg/L						
Duplicate (BDG0005-DUP1)			**Source: 22F2287-07**		Prepared: 7/1/2022 Analyzed: 7/7/2022					
Total Filterable Solids (TDS)	420		10.0	mg/L		427			1.57	5
Duplicate (BDG0005-DUP2)			**Source: 22F2287-10**		Prepared: 7/1/2022 Analyzed: 7/7/2022					
Total Filterable Solids (TDS)	420		10.0	mg/L		427			1.57	5
Reference (BDG0005-SRM1)					Prepared: 7/1/2022 Analyzed: 7/7/2022					
Total Filterable Solids (TDS)	310			mg/L	325.0		95.4	90-110		

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Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BDG0009										
Blank (BDG0009-BLK1)					Prepared: 7/1/2022 Analyzed: 7/6/2022					
Potassium	ND		0.500	mg/L						
Sodium	ND		1	mg/L						
Calcium	ND		0.1	mg/L						
Magnesium	ND		0.1	mg/L						
Blank (BDG0009-BLK2)					Prepared: 7/1/2022 Analyzed: 7/6/2022					
Sodium	ND		1	mg/L						
Potassium	ND		0.500	mg/L						
Calcium	ND		0.1	mg/L						
Magnesium	ND		0.1	mg/L						
LCS (BDG0009-BS1)					Prepared: 7/1/2022 Analyzed: 7/6/2022					
Potassium	34.8		0.500	mg/L	35.71		97.5	90-110		
Sodium	35		1	mg/L	35.71		98.6	90-110		
Calcium	36.3		0.1	mg/L	35.71		102	90-110		
Magnesium	36.4		0.1	mg/L	35.71		102	90-110		
LCS (BDG0009-BS2)					Prepared: 7/1/2022 Analyzed: 7/6/2022					
Calcium	37.2		0.1	mg/L	35.71		104	90-110		
Sodium	36		1	mg/L	35.71		100	90-110		
Potassium	35.4		0.500	mg/L	35.71		99.0	90-110		
Magnesium	37.1		0.1	mg/L	35.71		104	90-110		
Duplicate (BDG0009-DUP1)		**Source: 22F2283-03**			Prepared: 7/1/2022 Analyzed: 7/6/2022					
Calcium	38.0		0.1	mg/L		37.3			1.96	15
Potassium	2.28		0.500	mg/L		2.24			1.81	15
Sodium	69		1	mg/L		67			2.05	15
Magnesium	16.5		0.1	mg/L		16.2			1.47	15
Matrix Spike (BDG0009-MS1)		**Source: 22F2283-03**			Prepared: 7/1/2022 Analyzed: 7/6/2022					
Sodium	108		1	mg/L	35.71	67	114	90-110		
Potassium	38.2		0.500	mg/L	35.71	2.24	101	90-110		
Calcium	76.0		0.1	mg/L	35.71	37.3	108	90-110		
Magnesium	54.3		0.1	mg/L	35.71	16.2	107	90-110		
Matrix Spike (BDG0009-MS2)		**Source: 22F2287-08**			Prepared: 7/1/2022 Analyzed: 7/6/2022					
Potassium	36.8		0.500	mg/L	35.71	1.63	98.6	90-110		
Calcium	137		0.1	mg/L	35.71	94.3	120	90-110		
Sodium	185		1	mg/L	35.71	145	112	90-110		
Magnesium	89.4		0.1	mg/L	35.71	48.8	114	90-110		
Reference (BDG0009-SRM2)					Prepared: 7/1/2022 Analyzed: 7/6/2022					
Sodium	91			mg/L	91.50		99.1	90-110		

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Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BDG0009 (Continued)										
Reference (BDG0009-SRM2)					Prepared: 7/1/2022	Analyzed: 7/6/2022				
Potassium	21.4			mg/L	21.90		97.5	90-110		
Reference (BDG0009-SRM3)					Prepared: 7/1/2022	Analyzed: 7/6/2022				
Calcium	78.4			mg/L	73.90		106	90-110		
Magnesium	19.7			mg/L	19.30		102	90-110		

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Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BDG0010										
Blank (BDG0010-BLK1)					Prepared: 7/1/2022	Analyzed: 7/5/2022				
Kjeldahl Nitrogen (TKN), Total	ND		1.00	mg/L						
Total Nitrogen	ND		1.00	mg/L						
Blank (BDG0010-BLK2)					Prepared: 7/1/2022	Analyzed: 7/5/2022				
Kjeldahl Nitrogen (TKN), Total	ND		1.00	mg/L						
Total Nitrogen	ND		1.00	mg/L						
LCS (BDG0010-BS1)					Prepared: 7/1/2022	Analyzed: 7/5/2022				
Kjeldahl Nitrogen (TKN), Total	5.89		1.00	mg/L	5.709		103	90-110		
LCS (BDG0010-BS2)					Prepared: 7/1/2022	Analyzed: 7/5/2022				
Kjeldahl Nitrogen (TKN), Total	5.88		1.00	mg/L	5.709		103	90-110		
Duplicate (BDG0010-DUP1)		**Source: 22F2283-01**			Prepared: 7/1/2022	Analyzed: 7/5/2022				
Kjeldahl Nitrogen (TKN), Total	ND		1.40	mg/L		ND				10
Duplicate (BDG0010-DUP2)		**Source: 22F2287-06**			Prepared: 7/1/2022	Analyzed: 7/5/2022				
Kjeldahl Nitrogen (TKN), Total	ND		1.40	mg/L		ND				10
Matrix Spike (BDG0010-MS1)		**Source: 22F2283-01**			Prepared: 7/1/2022	Analyzed: 7/5/2022				
Kjeldahl Nitrogen (TKN), Total	8.48		1.40	mg/L	7.992	ND	106	90-110		
Matrix Spike (BDG0010-MS2)		**Source: 22F2287-06**			Prepared: 7/1/2022	Analyzed: 7/5/2022				
Kjeldahl Nitrogen (TKN), Total	8.25		1.40	mg/L	7.992	ND	103	90-110		
Reference (BDG0010-SRM1)					Prepared: 7/1/2022	Analyzed: 7/5/2022				
Kjeldahl Nitrogen (TKN), Total	11.1			mg/L	10.30		108	90-110		

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

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DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 · 800 228-9896 · Fax 559 268-8174

No. Samples: 10 Material: Water

Purchase Order No | Bill To: Acct # 23620 | Cons # 08 ~~09~~ SD 6/30/22

Name: Bar Mac Dairy/5N Cattle
Address: 5070 S. Hunt Rd.
City: Gustine **State:** CA **Zip:** 95322
Telephone: **Fax:**
Cell/Email: dejagerfarms@gmail.com
COPY TO: ariordan@fragservices.com dustin@dejagerfarms.com
REQUESTED BY: Brad Nightengale
RANCH: BM/5n
ID CROP:
Present Stage of Growth
Intended Previous
Date Sampled: 6/29 **Sampled By:** DJF

Analysis Required:

( ) Manure: DM1 (N,P,K, Moisture)
( ) Manure: DM2 (DM1 Plus Ash, Ca, Mg, Na, S, Cl)
( ) Soil Spring: DS1 (NO_3-N (0'-1')
( ) Soil Fall: DS2 (FA1 (0-1'); NO_3-N (1-2'))
( ) Harvest Tissue: CRA1 (N, P, K, Moisture, Ash)
( ) Feed: ADF/NDF
( ) Petiole/Midrib: P1
( ) Leaf: L1
(X) Other: DWWZ plus TN

	DESCRIPTION OF SAMPLE	Temp
1	BM - Dom milk Barn	-0.7
2	BM - Dom Dairy Barn	1.6
3	BM - Dom Hay Barn	2.1
4	BM - IW Hunt	0.3
5	BM - IW # 7	0.4
6	BM - IW # 24	0.1
7	BM - CCID Canal	0.9
8	5n - Dom 1	0.0
9	5n - Dom 2	0.6
10	5n - Diesel Well	3.5
11		
12		
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23		
24		

[X] QA/QC Document [X] Copy of Chain [ ] Comments/Recommendations

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	JLF	DLL	63022 13:03	
Second			6/30 1343	
Third				
Fourth				

I guarantee that as the client, or on behalf of client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a liquidated damage fee of 2% per month (annually 24%) or $5.00 per month whichever is greater.
If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc. (*cal*). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through *cal* under its Rules and Procedures. The parties will equally bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all mediation and arbitration costs, and in the event of arbitration, reasonable attorneys' fees of Dellavalle Laboratory.

Invoicing Information: ***Shipping***
Sampling hrs $ In
Miles $ Out
Consulting

Amt Paid	Rec By	Check #	Date

Signature
Sample received in cooler with ice (coolant)
[ ] Yes [ ] No

X:\1-CLIENT CUSTOM FIELDSHEETS\FIELDSHEETS\Dairy Fieldsheets\[F & R Ag Services (Alex).xls]Soil & Tissue rev 4.18.18

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Shipping Information: Shipped In ☐ Picked-Up ☑ Walk In ☐ DLI Sampler ☐ Other ☐

Container: Ice Chest ☑ Box ☐ None ☐ **Refrigerant:** Wet Ice ☑ Blue Ice ☐ None ☐

Samples Preserved with HNO_3 or H_2SO_4 were: ☐ Received Preserved ☑ Preserved Upon Receipt at Laboratory

Type of Container(s) Received		Sample Number 1	2	3	4	5	6	7	8	9	10
Sample Containers for Internal (DLI) Use *(Containers that go into the Lab)*											
Plastics	100 mL sterile plastic $Na_2S_2O_3$ (Green)										
Plastics	250 mL unpreserved (White) Plastic										
Plastics	250 mL HNO_3 (Red) Plastic										
Plastics	* pH Value										
Plastics	250 mL H_2SO_4 (Yellow) Plastic										
Plastics	* pH Value	<2	<2	<2	<2	<2	<2	<2	<2	<2	<2
Plastics	500 mL unpreserved (White) Plastic										
Plastics	1 L unpreserved (White) Plastic	1	1	1	1	1	1	1	1	1	1
Plastics	1 L unpreserved (BOD) (Purple) Plastic										
Special	500mL unpreserved (White) Glass										
Special	PO4-P Kit										
Special	**Other:**										
Sample Containers for Subcontracted ("Send Out") Analyses *(Containers that go in the Subcontract ("Send Out") Refrigerator)*											
Plastics	100 mL sterile plastic $Na_2S_2O_3$ (Green)										
Plastics	250 mL unpreserved (White) Plastic										
Plastics	250 mL HNO_3 (Red) Plastic										
Plastics	250 mL H_2SO_4 (Yellow) Plastic										
Plastics	500 mL HNO_3 (Red)										
Plastics	1 L unpreserved (White) Plastic										
Plastics	1 L unpreserved (BOD) (Purple) Plastic										
Plastics	1 L HNO_3 (Red)										
VOA Vials	40 mL VOA, $Na_2S_2O_3$ + MCAA (EPA531)										
VOA Vials	40 mL VOA, $Na_2S_2O_3$ (EPA547)										
VOA Vials	40mL AG VOA unpreserved (White) (Set of 3)										
VOA Vials	40 mL AG VOA, $Na_2S_2O_3$ (Green) (Set of 3)										
VOA Vials	40mL VOA, H_3PO_4 (Set of 3)										
VOA Vials	40 mL VOA, HCl (Blue) (Set of 3)										
VOA Vials	40 mL VOA, $Na_2S_2O_3$ (Green) (Set of 3)										
Glass	250 mL AG unpreserved (White)										
Glass	250 mL AG H_2SO_4 (Yellow)										
Glass	250 mL AG $Na_2S_2O_3$ (Green)										
Glass	250 mL AG $Na_2S_2O_3$ + MCAA										
Glass	500 mL glass unpreserved (White)										
Glass	500 mL AG HCl (Blue)										
Glass	1 L AG unpreserved (White)										
Glass	1 L AG H_2SO_4 (Yellow)										
Glass	1 L AG $Na_2S_2O_3$ (Green)										
Glass	1 L AG HCl (Blue)										
Special	Cr^{6+} - 50mL Plastic w/Borate/HCO_3/CO_3										
Special	Cyanide - 500 mL NaOH										
Special	Asbestos - 1L P wrapped in foil (Set of 2)										
Special	Sulfide - 1 L **AG** or **P** NaOH + ZnAc										
Special	Chlorite/Bromate - 250 mL AG with EDA										
Special	HAA5 - 250mL AG Ammonium Chlorite										
Special	DO KIT										
Special	**Other:**										
Special	**Other:**										